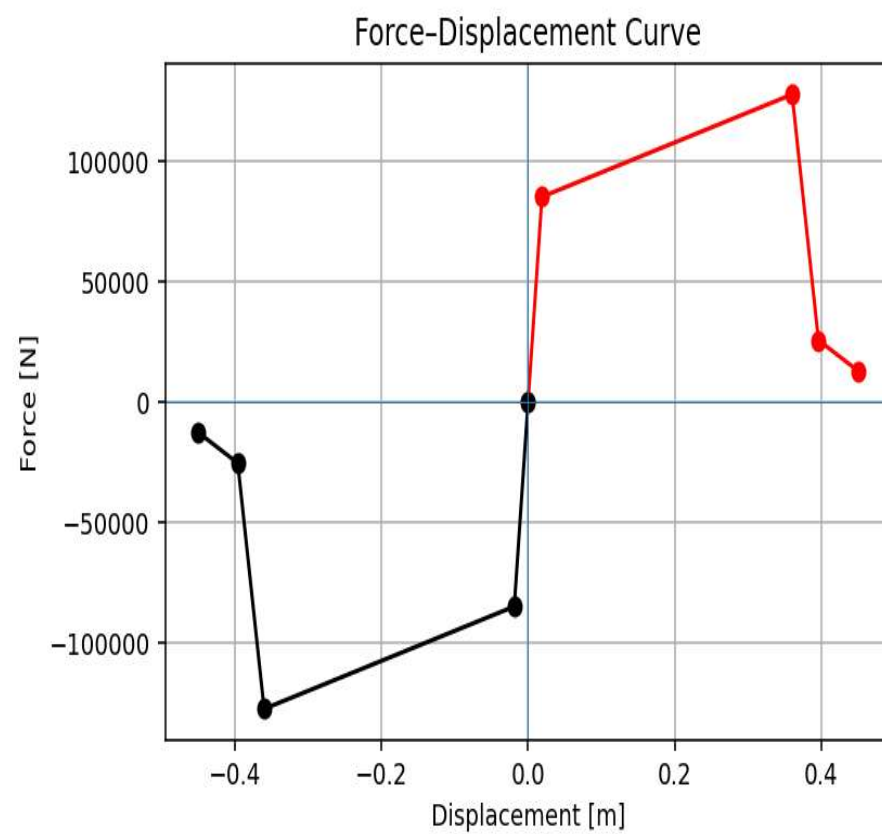
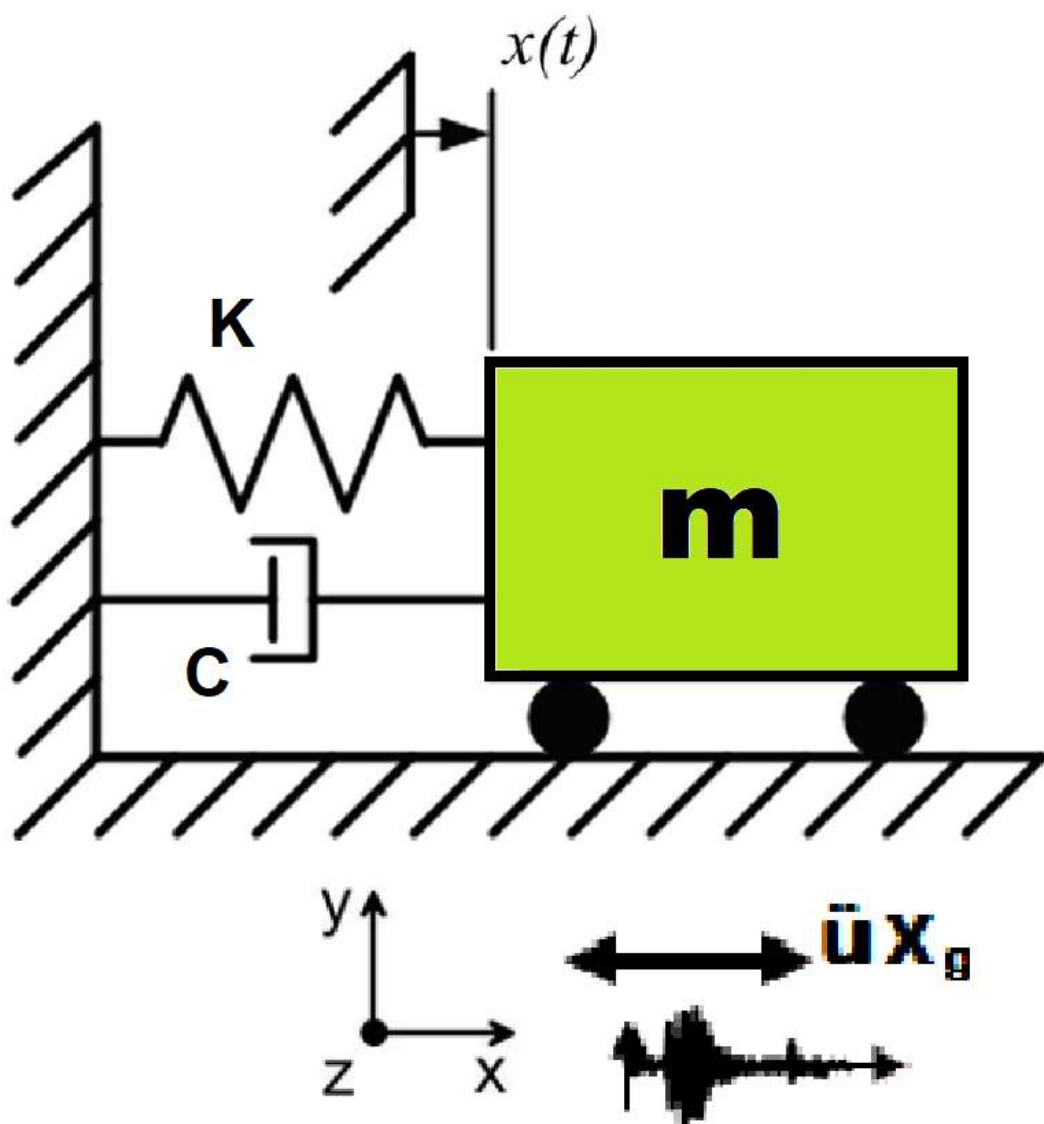


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

**FRAGILITY ANALYSIS BASED ON
ACCELERATION, STRUCTURAL DUCTILITY
DAMAGE INDEX, ENERGY DISSIPATION
CAPACITY INDEX, EQUIVALENT VISCOUS
DAMPING RATIO WITH NONLINEAR DYNAMIC
ANALYSIS OF A SINGLE-DEGREE-OF-FREEDOM
(SDOF) SYSTEM UTILIZING 200 GROUND
MOTIONS IN OPENSEES**

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

Δ_u = Lateral Ultimate Displacement from Pushover Analysis

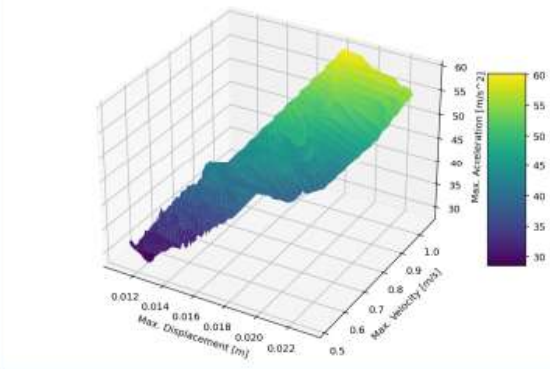
FileEditSearchSourceRunDebugConsolesProjectsToolsViewHelp

SDOF_SEISMIC_EDCI_EVDR.py

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # FRAGILITY ANALYSIS BASED ON ACCELERATION, STRUCTURAL DUCTILITY DAMAGE INDEX, ENERGY DISSIPATION
4 # CAPACITY INDEX, EQUIVALENT VISCOUS DAMPING RATIO WITH NONLINEAR DYNAMIC ANALYSIS
5 # OF A SINGLE-DEGREE-OF-FREEDOM (SDOF) SYSTEM UTILIZING 200 GROUND MOTIONS IN OPENSEES
6 # -----
7 # INELASTIC SEISMIC ANALYSIS WITH 200 GROUND MOTIONS
8 # -----
9 # EQUIVALENT VISCOUS DAMPING RATIO: xi_eq = 100 * E_d / (4 * pi * E_s)
10 # -----
11 # ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacement)
12 # -----
13 # This program performs Dynamic Analysis on a Single-Degree-of-Freedom (SDOF) system
14 # subjected to 200 seismic ground motions. The analysis evaluates the structural response under varying
15 # levels of seismic intensity.
16 # The framework is designed to support researchers and engineers in assessing the probabilistic seismic
17 # performance of structures, with a focus on understanding the impact of uncertainty on structural
18 # response and design.
19 # -----
20 # Key Features:
21 # - Simulation of SDOF system using OpenSees.
22 # - Incremental scaling of ground motions for Nonlinear Dynamic Analysis.
23 # - Probabilistic fragility assessment based on predefined damage states.
24 # - Visualization of structural response and fragility curves.
25 # - Export of results for further analysis.
26 # -----
27 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
28 # EMAIL: salar.d.ghashghaei@gmail.com
29 #####
30
31 """
32 This code implements a comprehensive nonlinear dynamic analysis framework for
33 performance-based earthquake engineering assessment of single-degree-of-freedom
34 (SDOF) systems. The methodology combines traditional nonlinear time-history
```

32 %

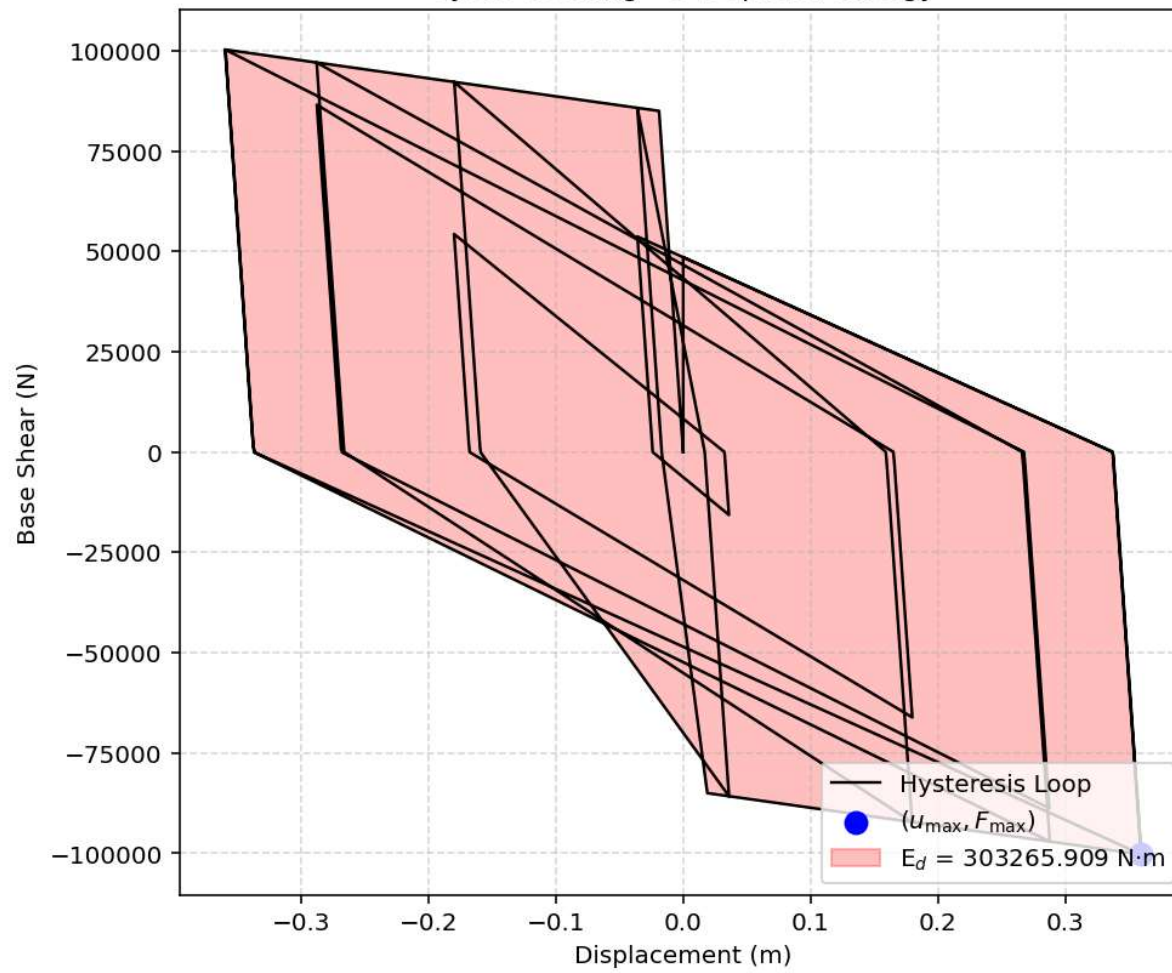
3D CONTOUR: Max. Displacement [m] - Max. Velocity [m/s] - Max. Acceleration [m/s^2]

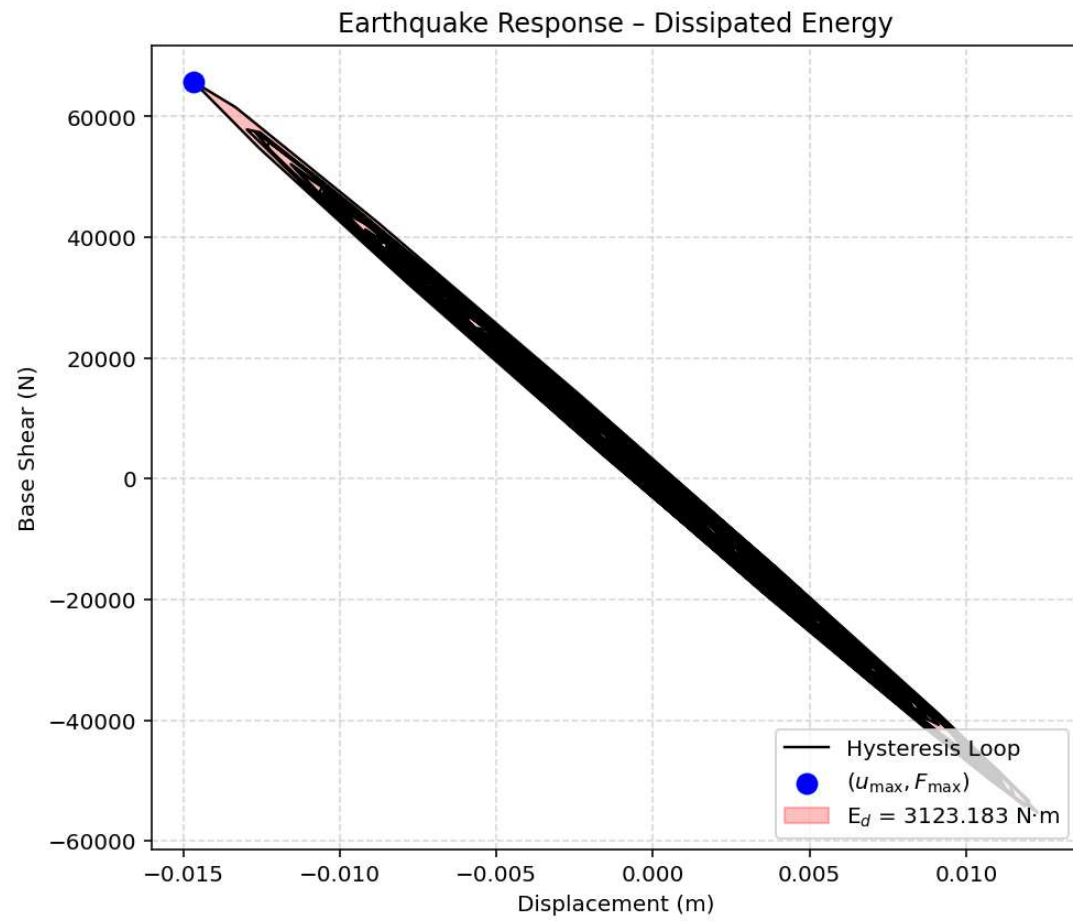


IPython ConsolePlotsVariable ExplorerHelpDebuggerHistoryFiles

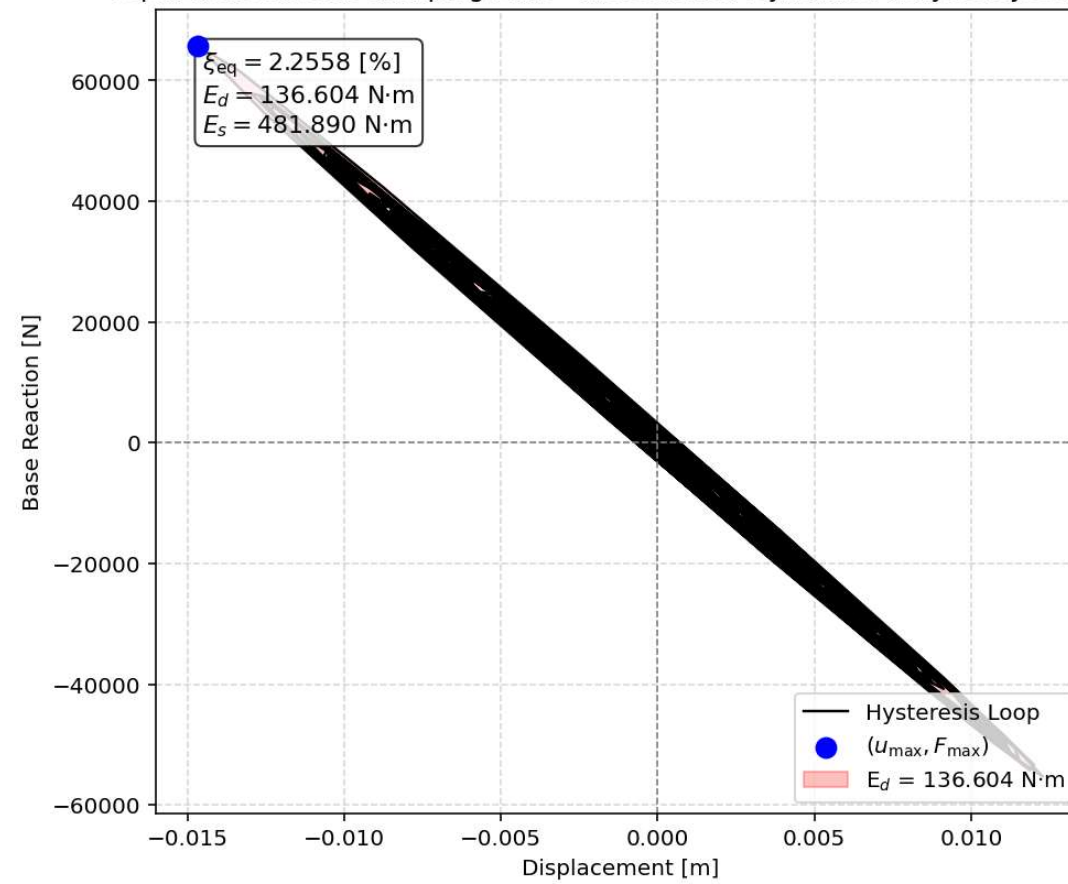
InlineConda: anaconda3 (Python 3.12.7) ✓ LSP: PythonLine 14, Col 106UTF-8CRLFRWMem 82%

Cyclic Loading - Dissipated Energy

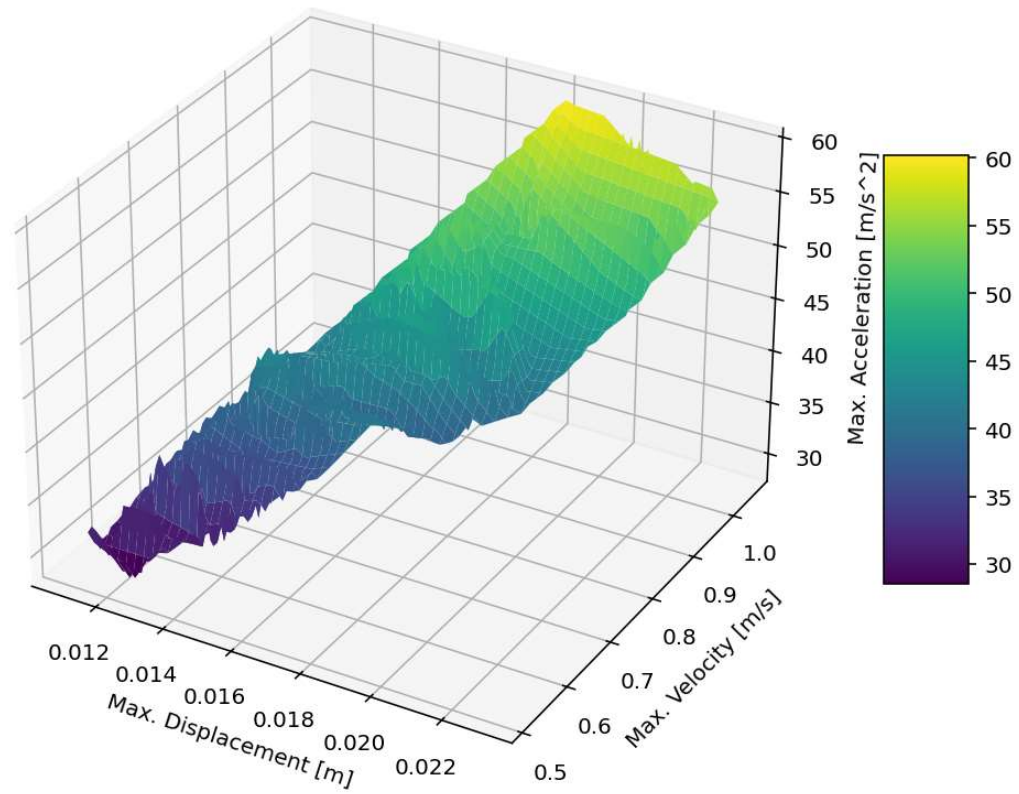


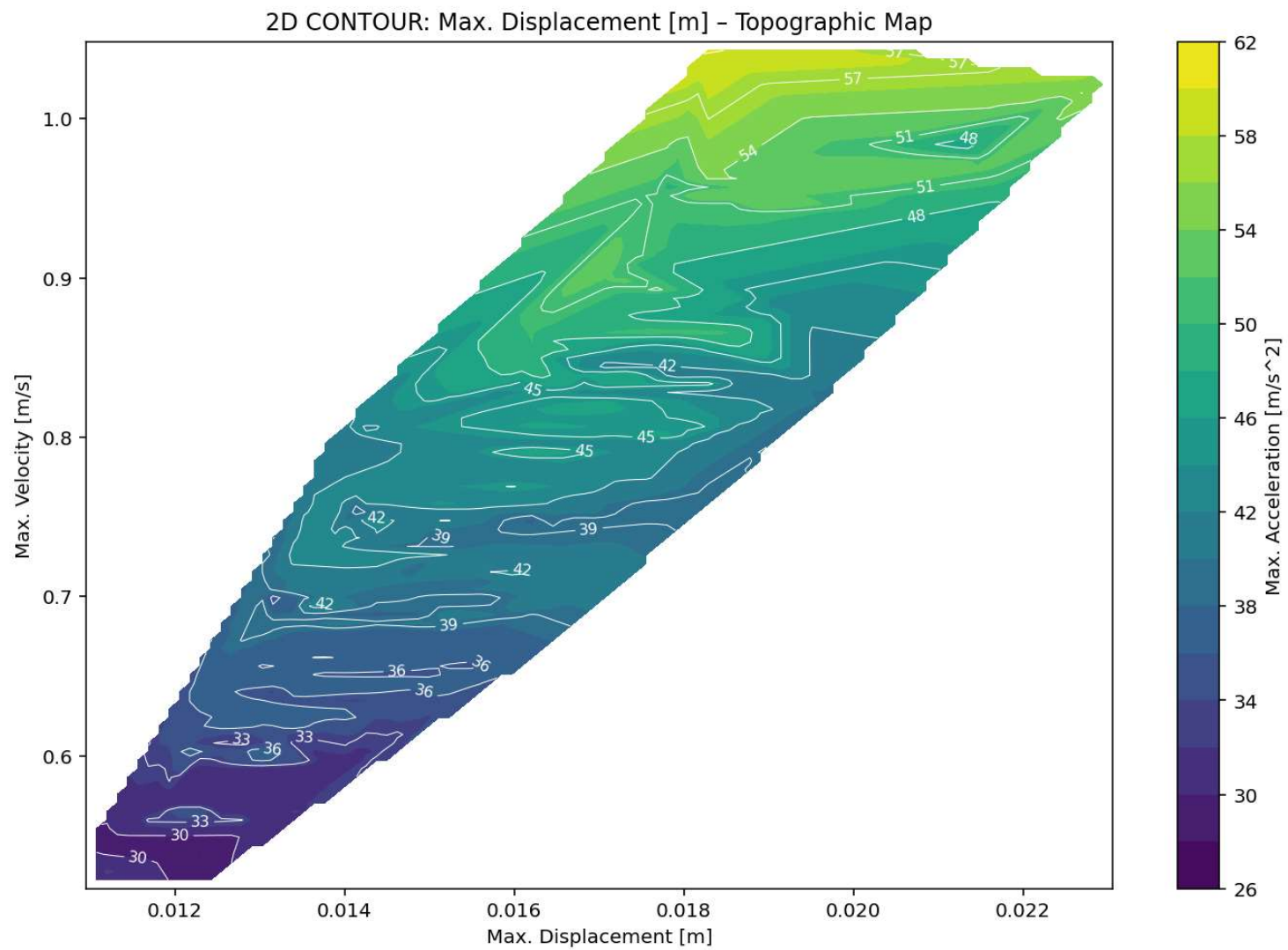


Equivalent viscous damping ratio - Incremental Dynamic Analysis Hysteresis

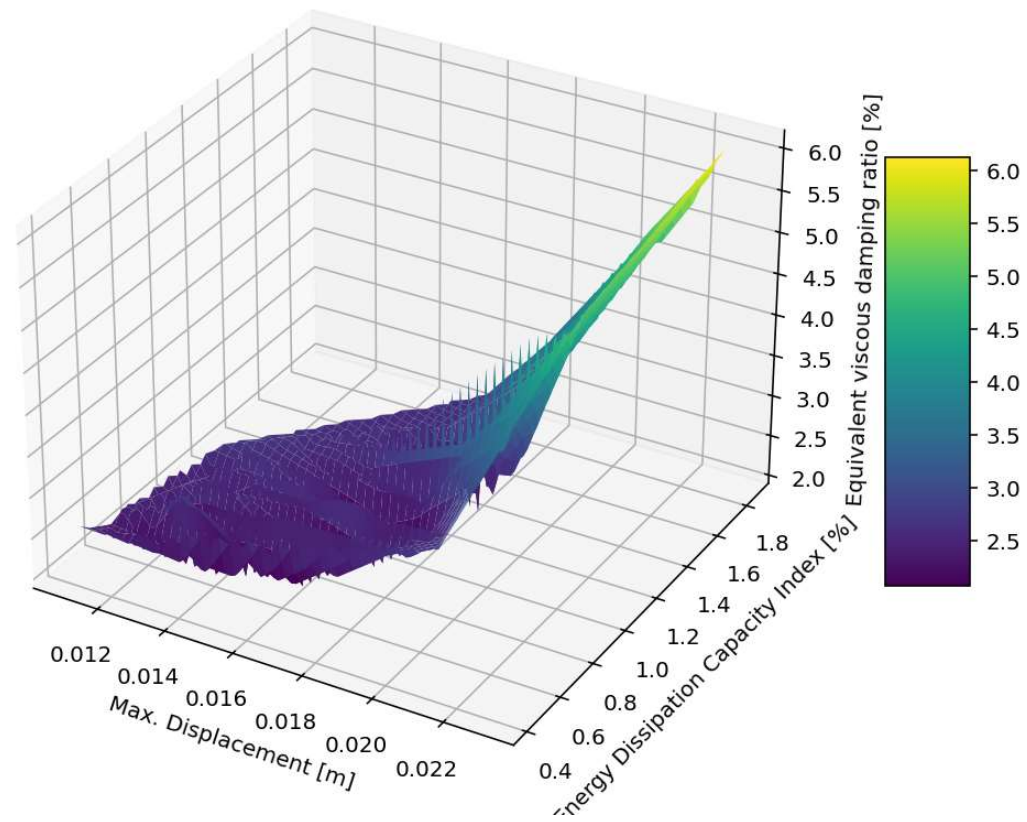


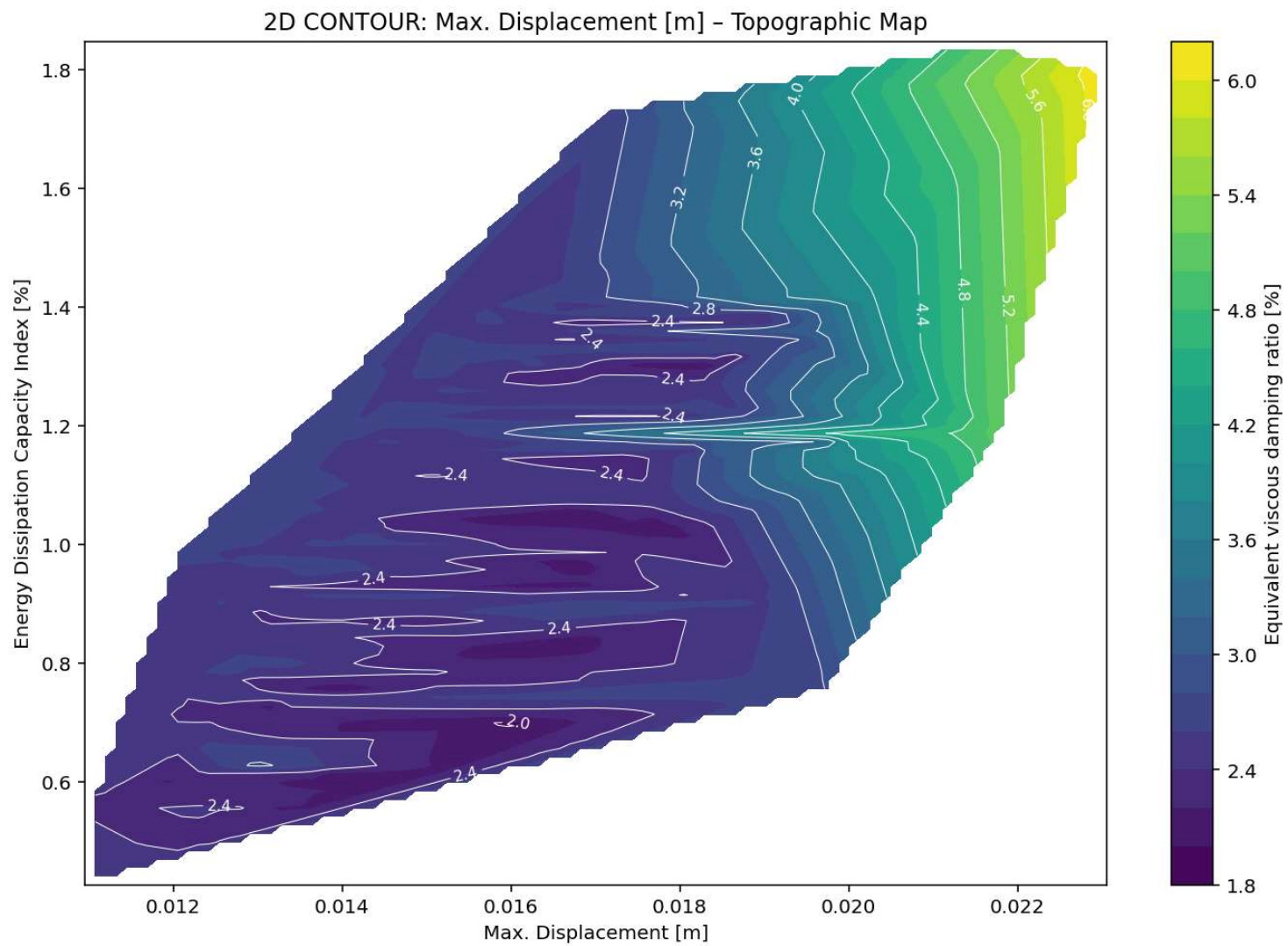
3D CONTOUR: Max. Displacement [m] - Max. Velocity [m/s] - Max. Acceleration [m/s²]



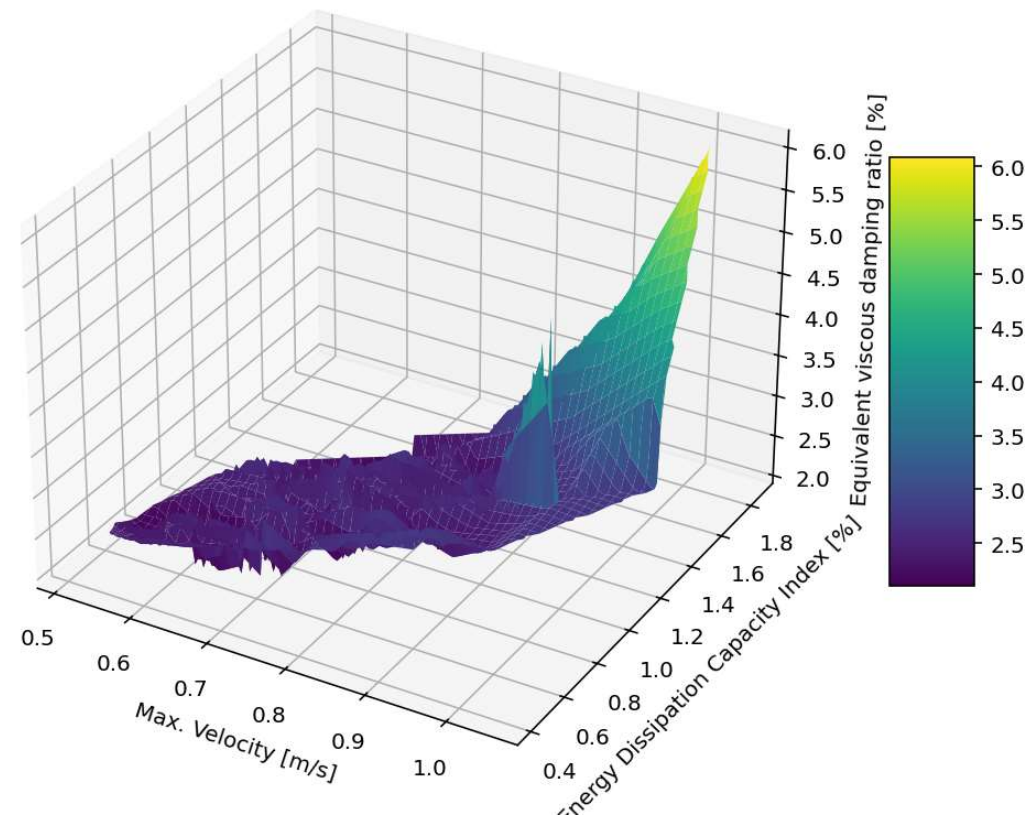


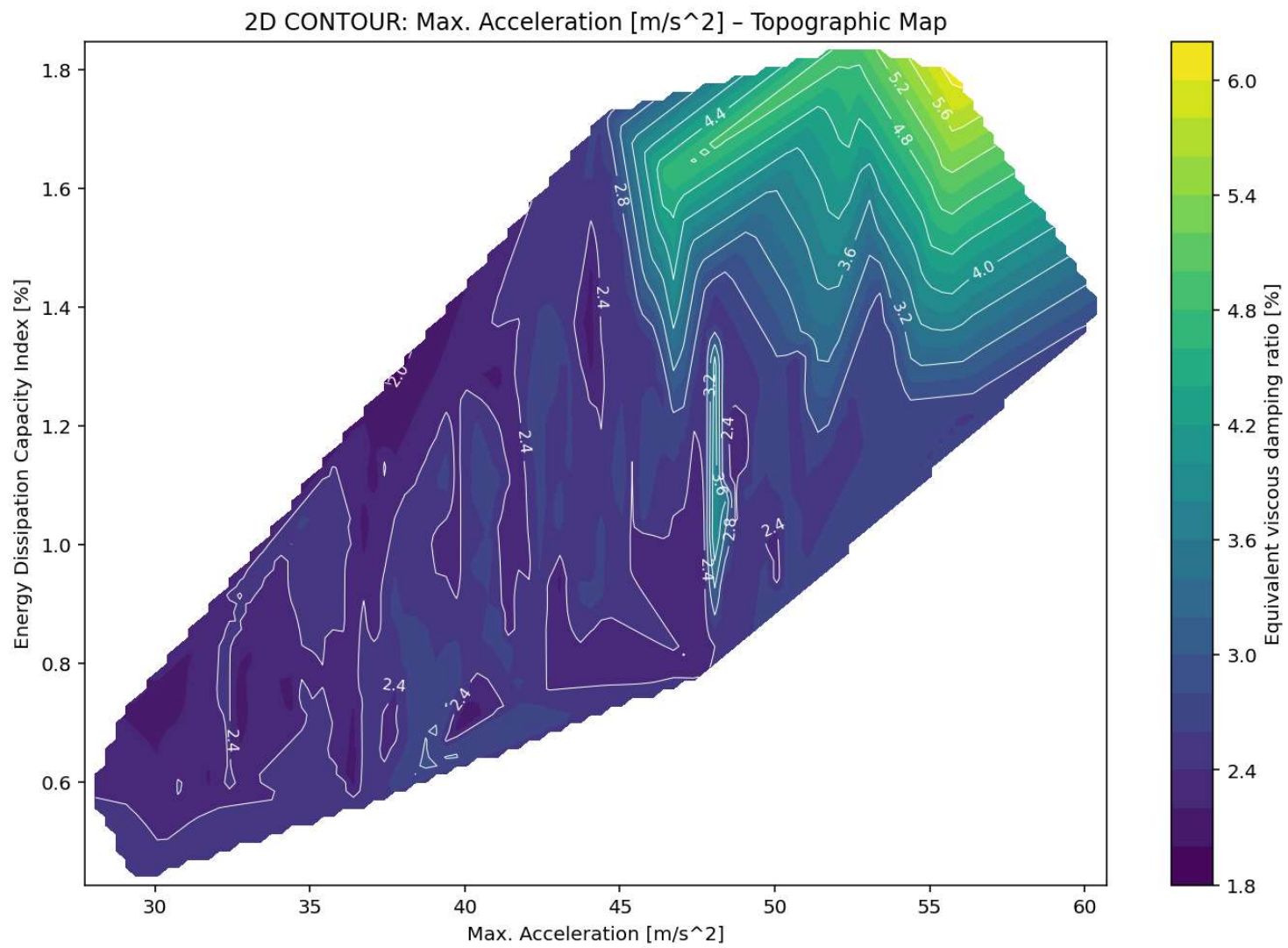
3D CONTOUR: Max. Displacement [m] - Energy Dissipation Capacity Index [%] - Equivalent viscous damping ratio [%]





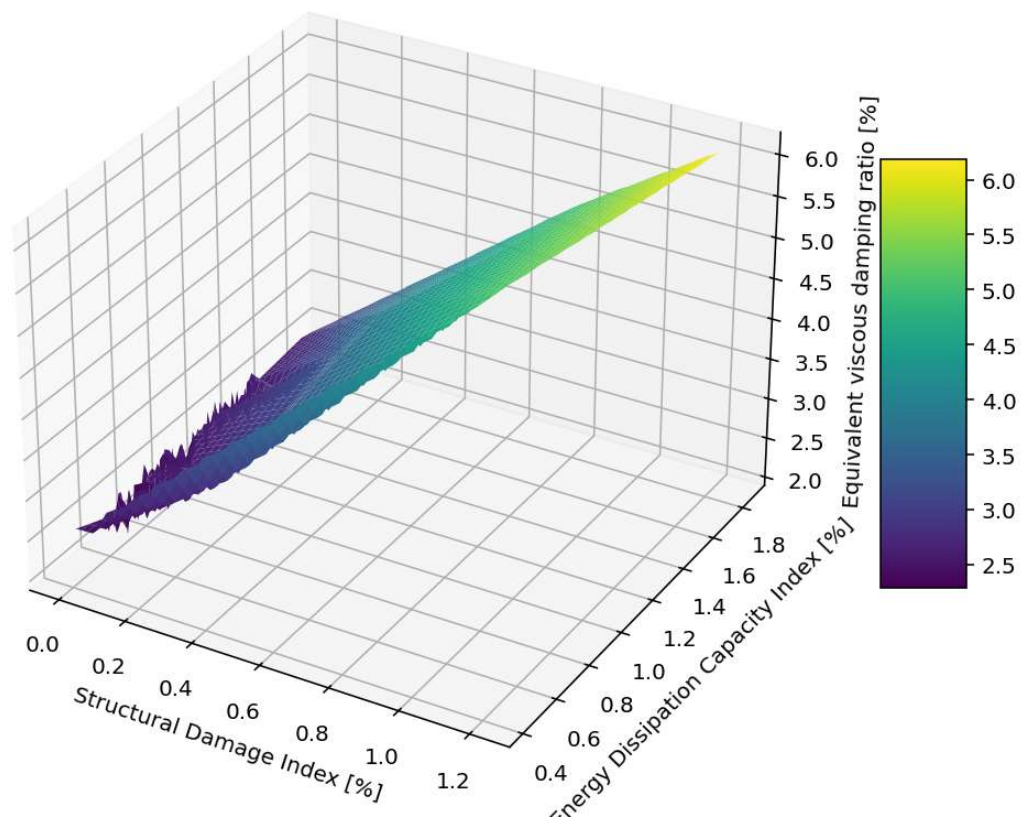
3D CONTOUR: Max. Velocity [m/s] - Energy Dissipation Capacity Index [%] - Equivalent viscous damping ratio [%]



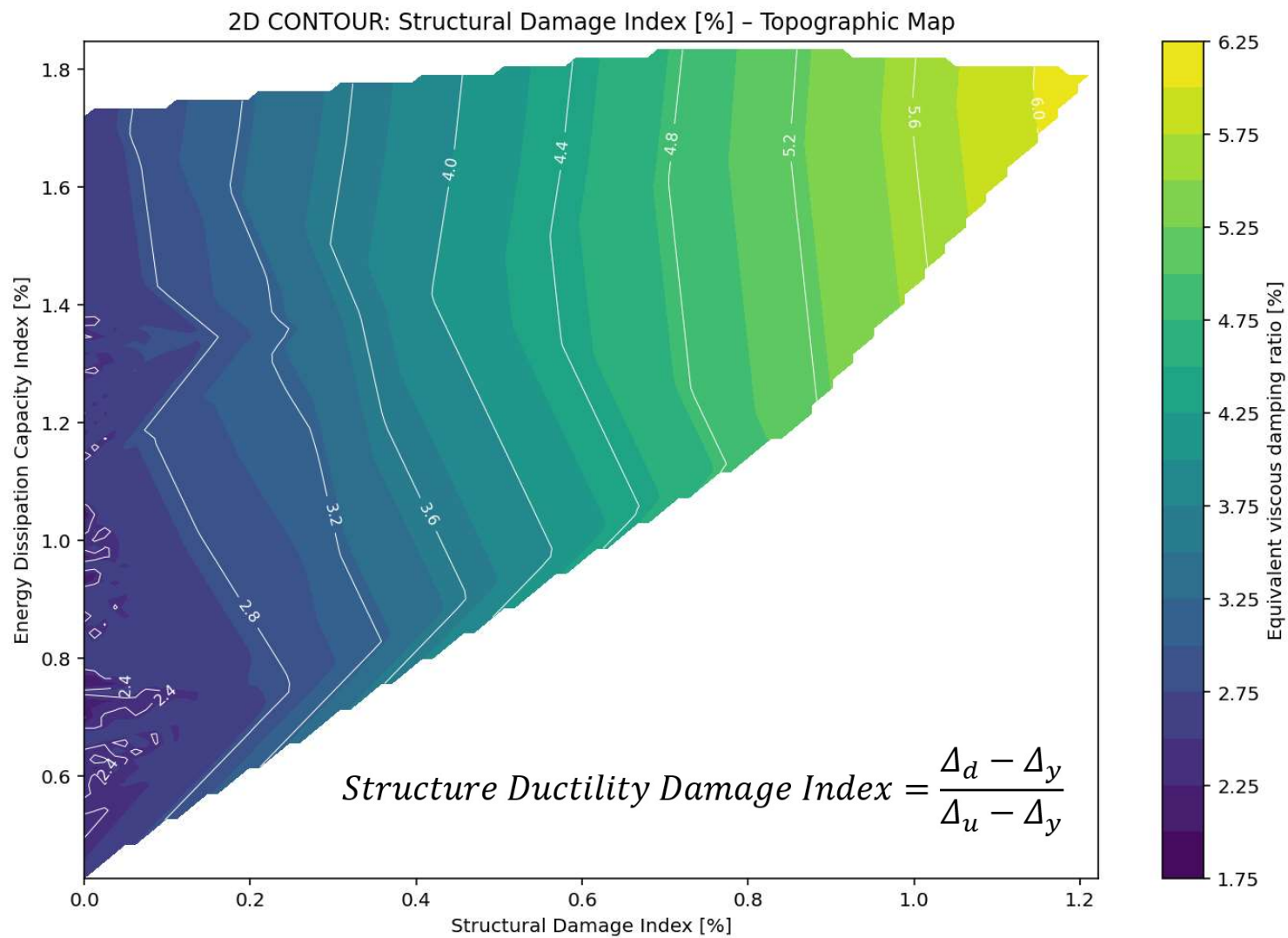


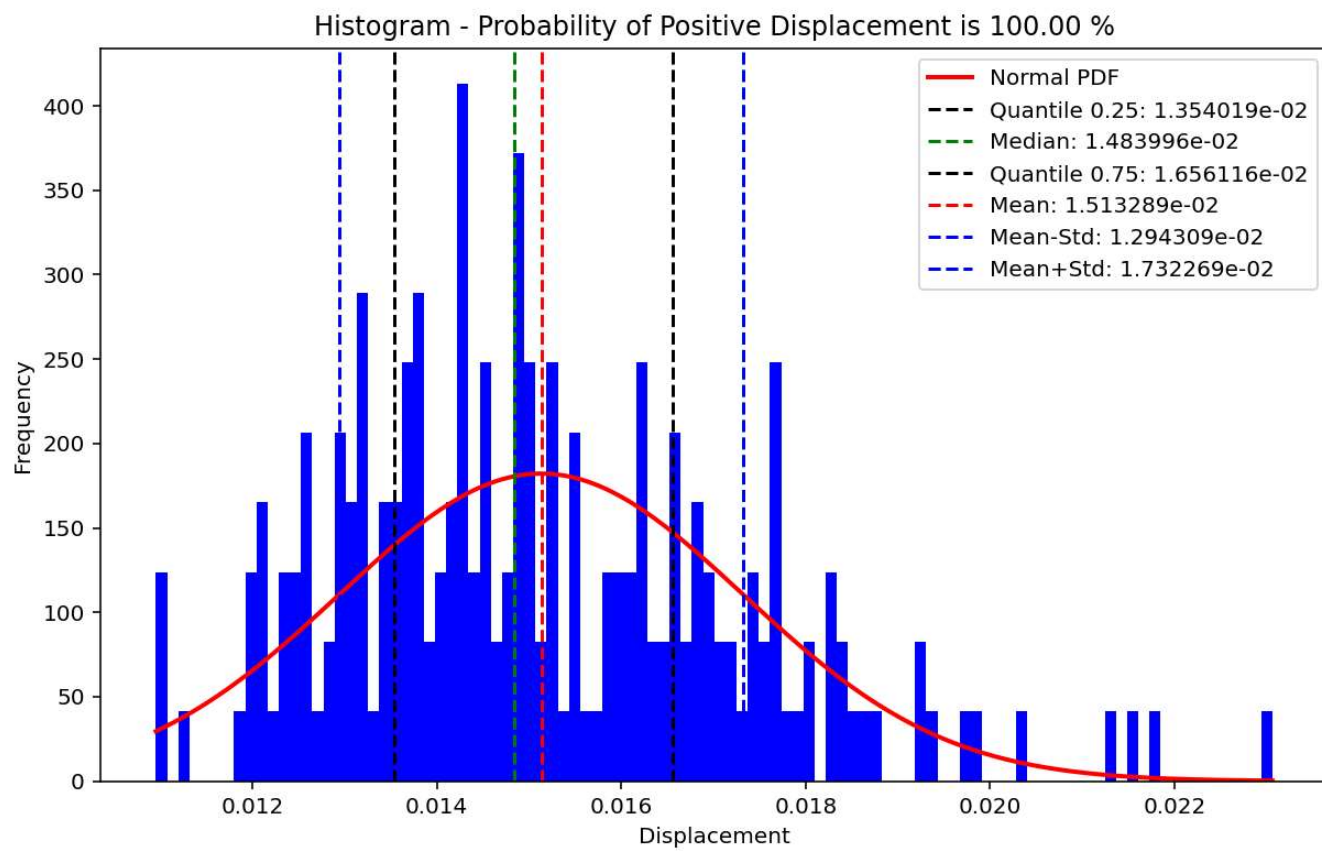
3D CONTOUR: Structural Damage Index [%] - Energy Dissipation Capacity Index [%] - Equivalent viscous damping ratio [%]

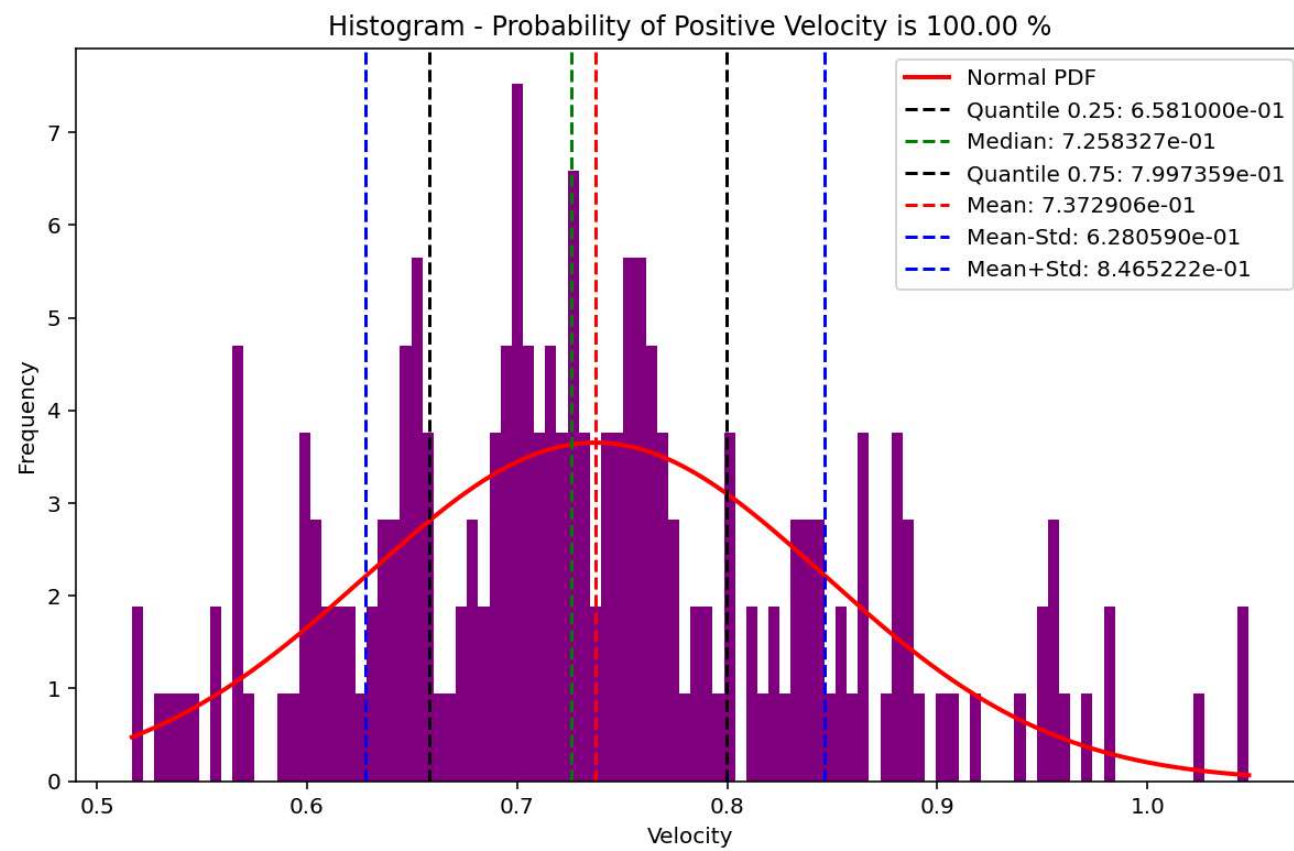
$$\xi_{eq} = \frac{E_d}{4\pi E_s} * 100$$

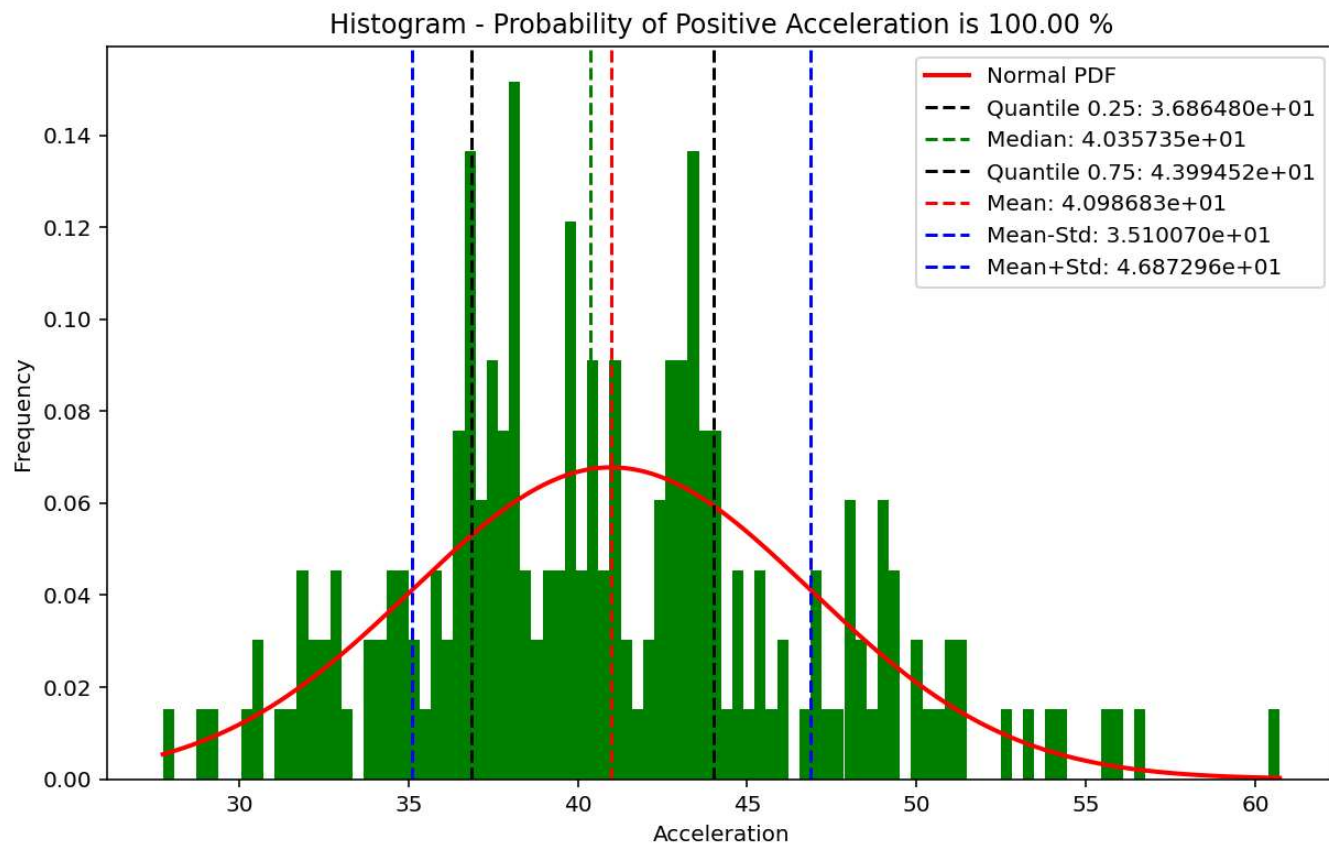


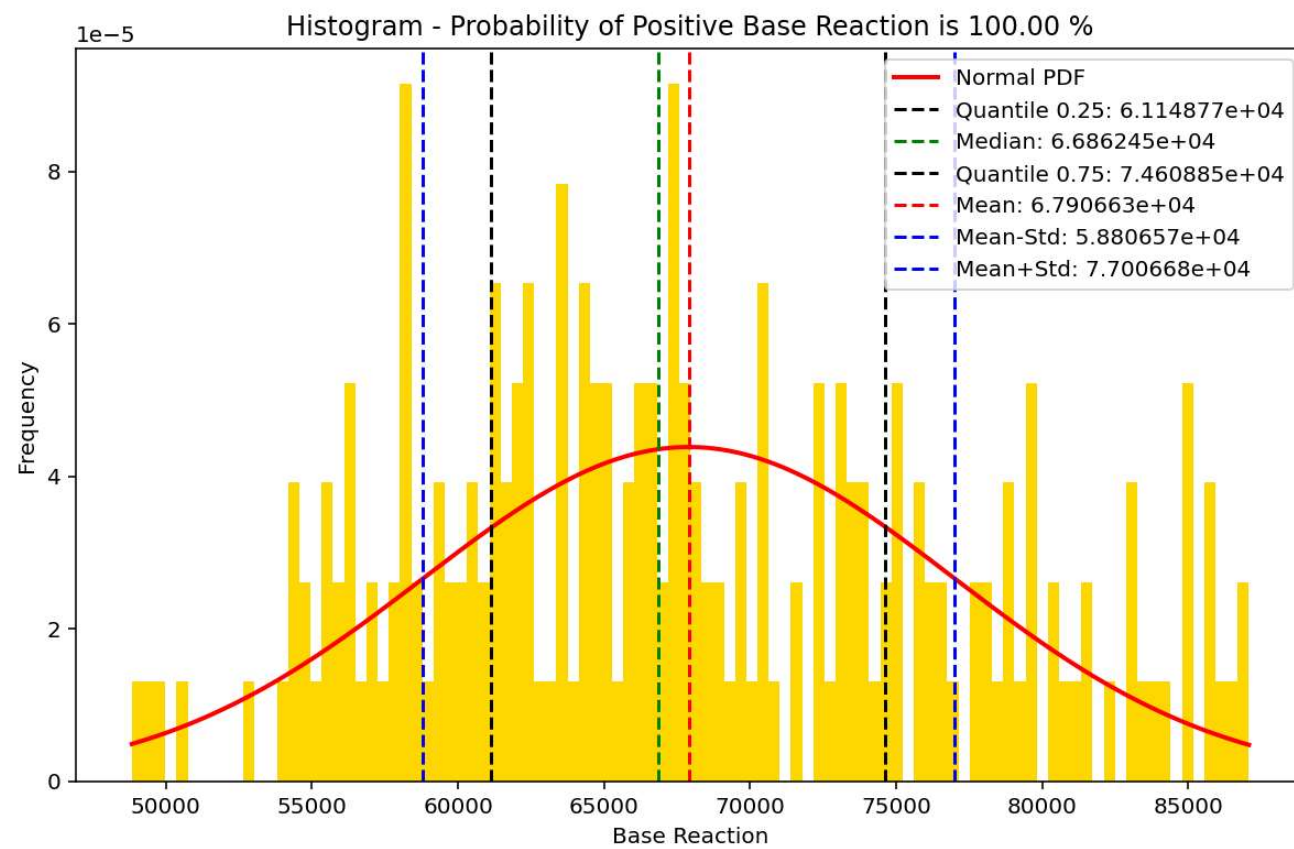
$$\text{Energy Dissipation Capacity Index} = \frac{E_d(\text{Dissipated Energy from Earthquake})}{E_d(\text{Dissipated Energy from Cyclic Displacement})} * 100$$

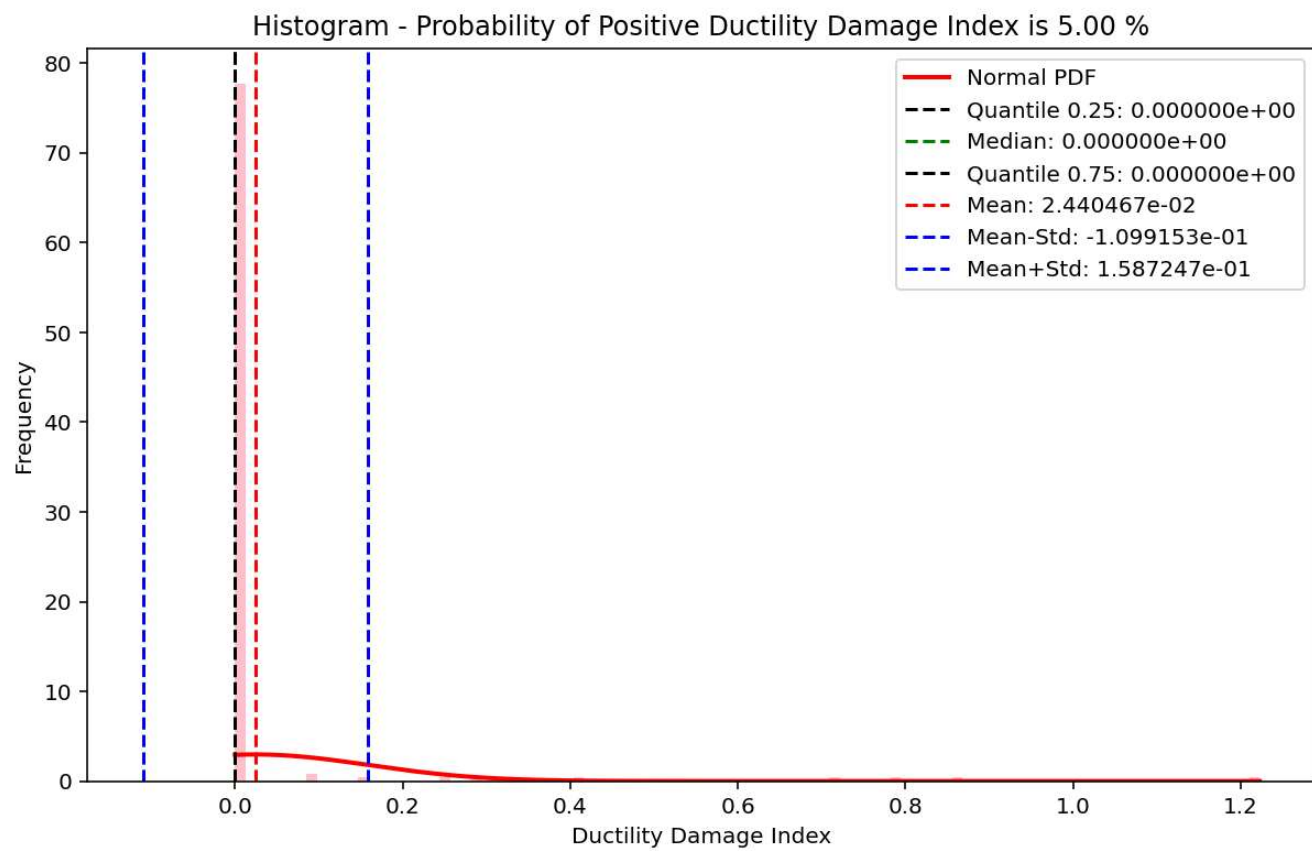


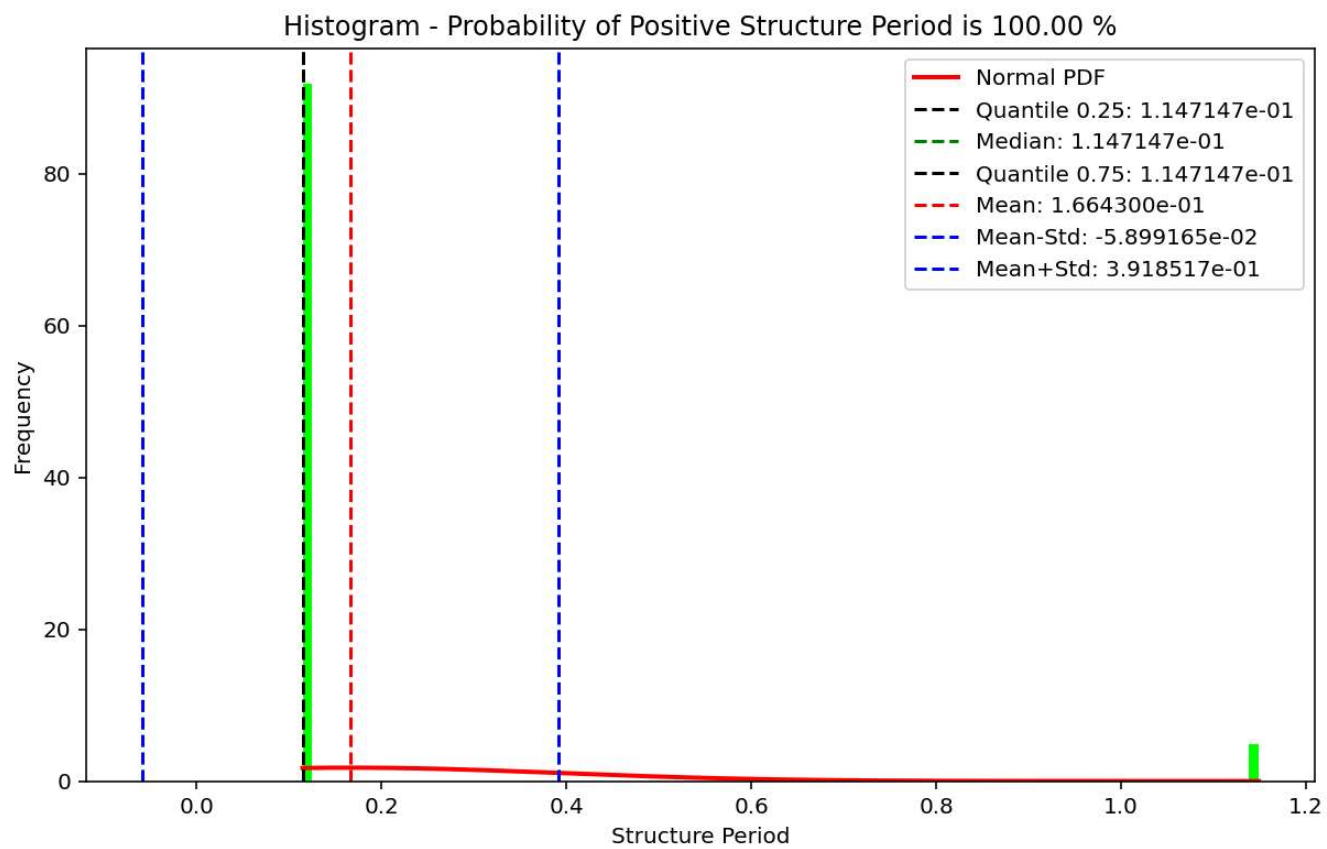


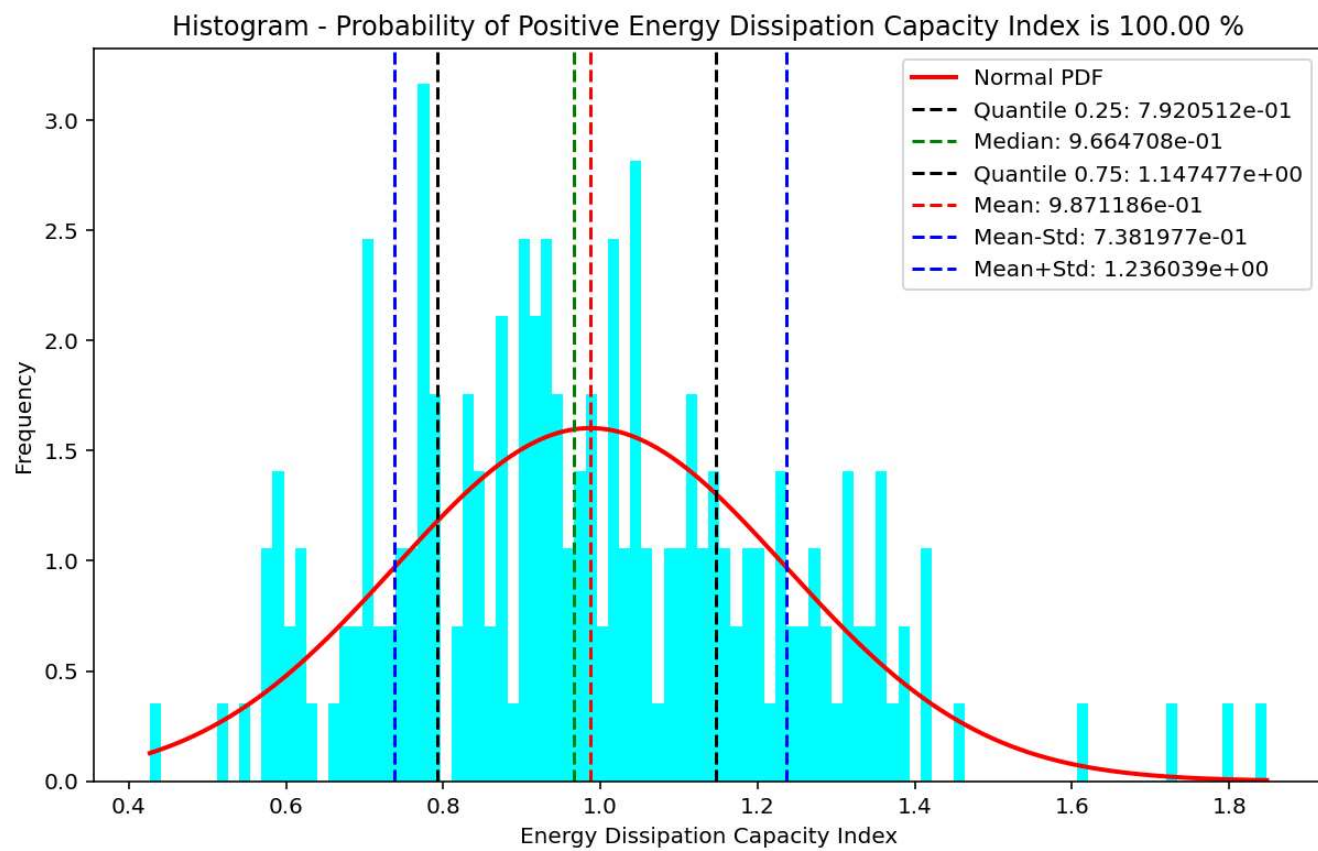


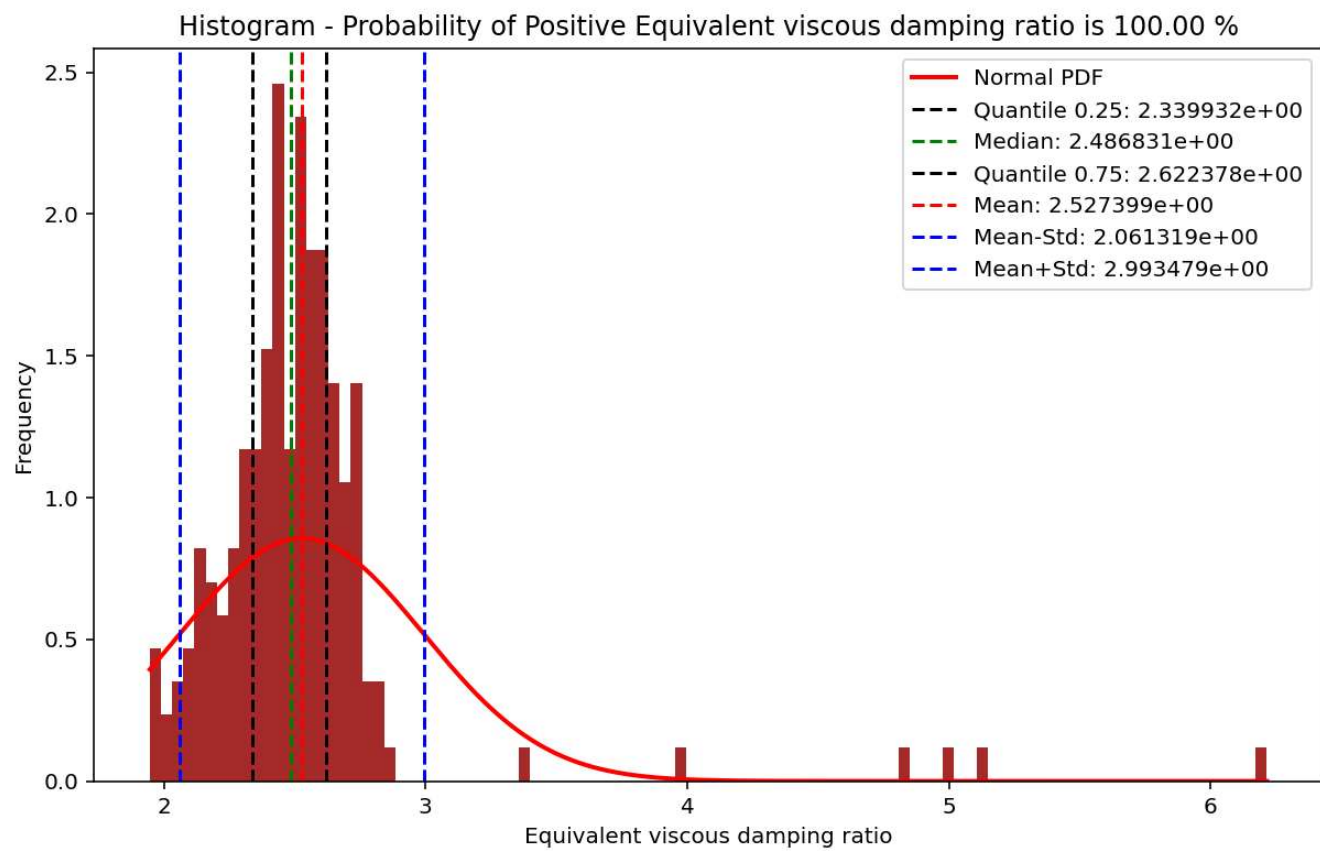


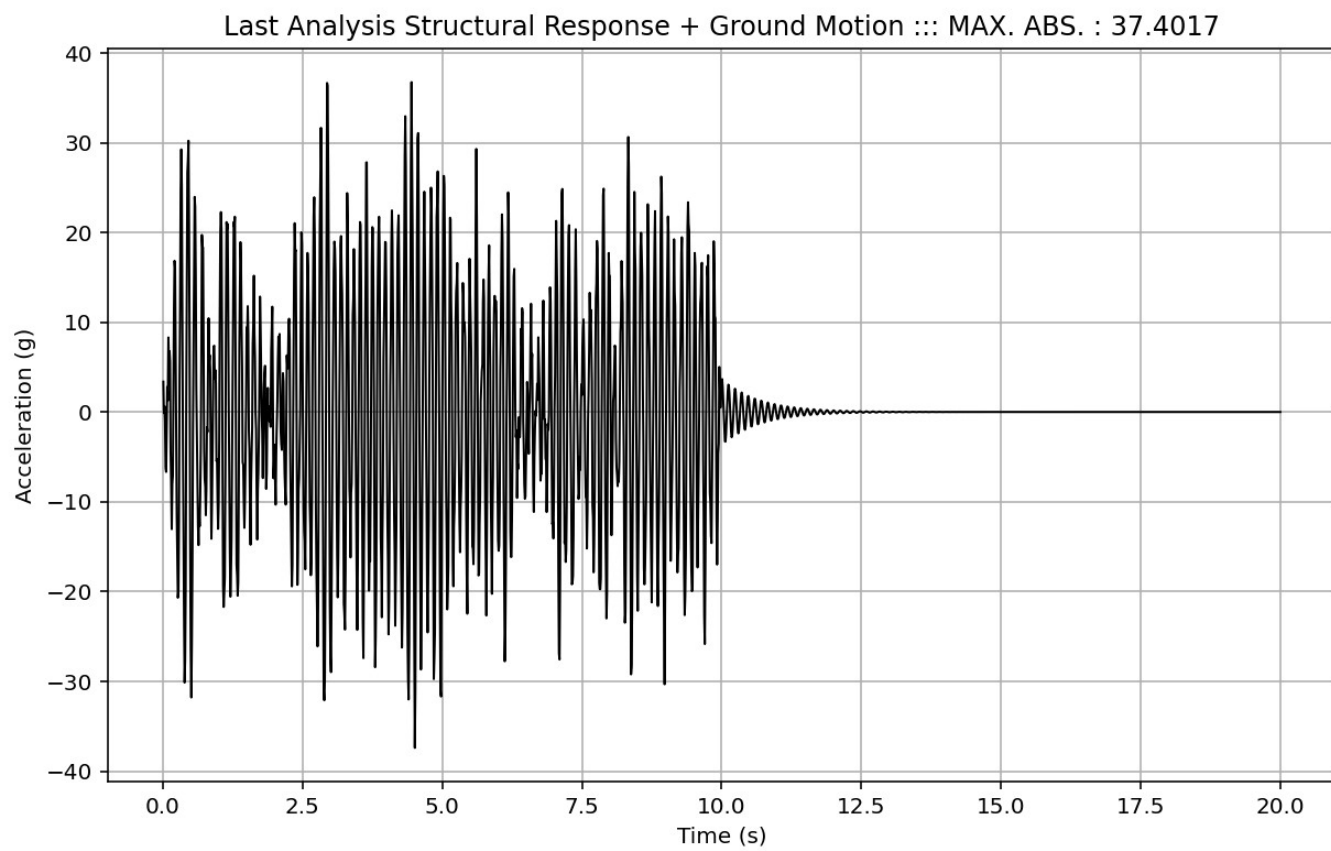


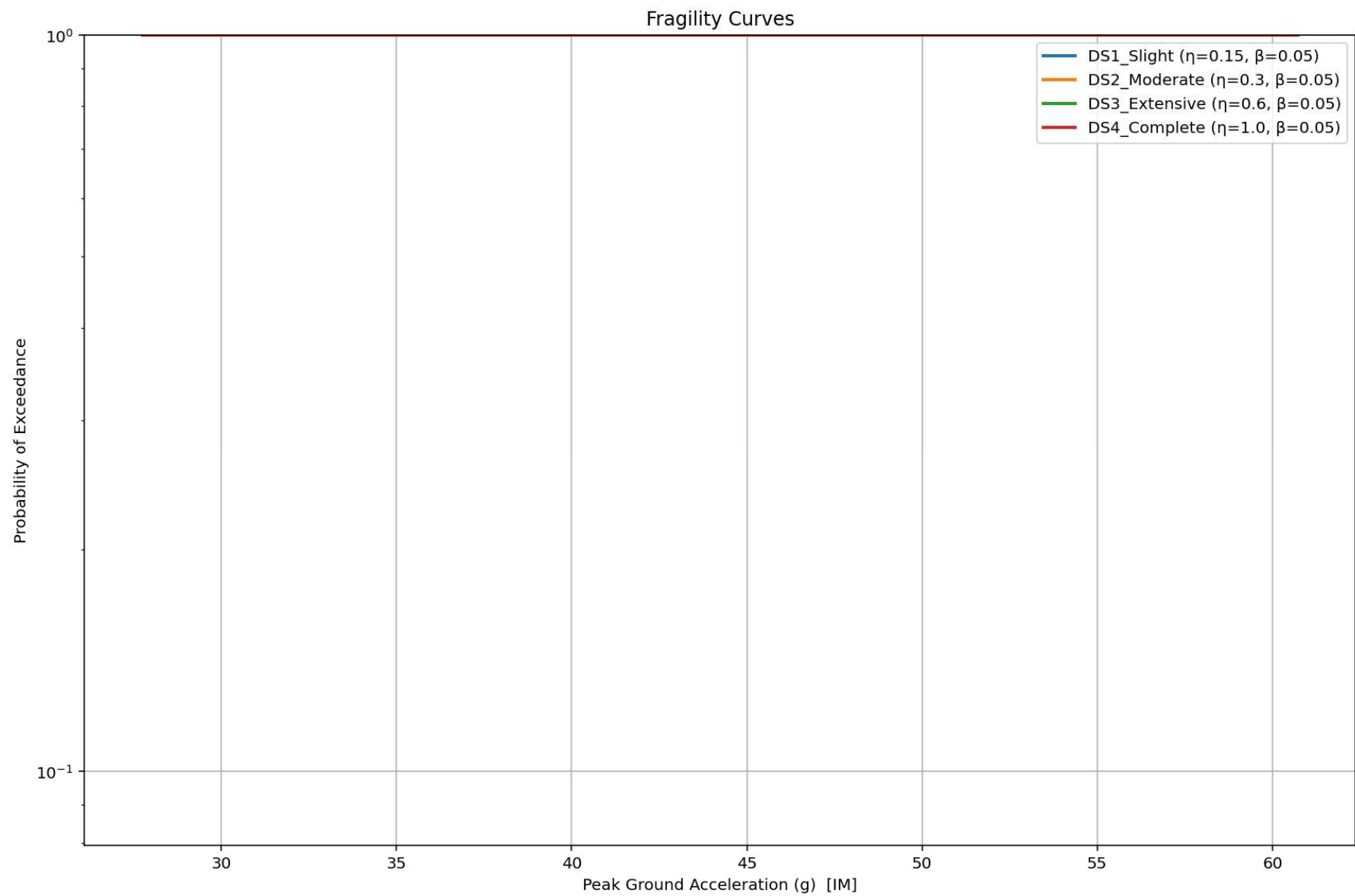




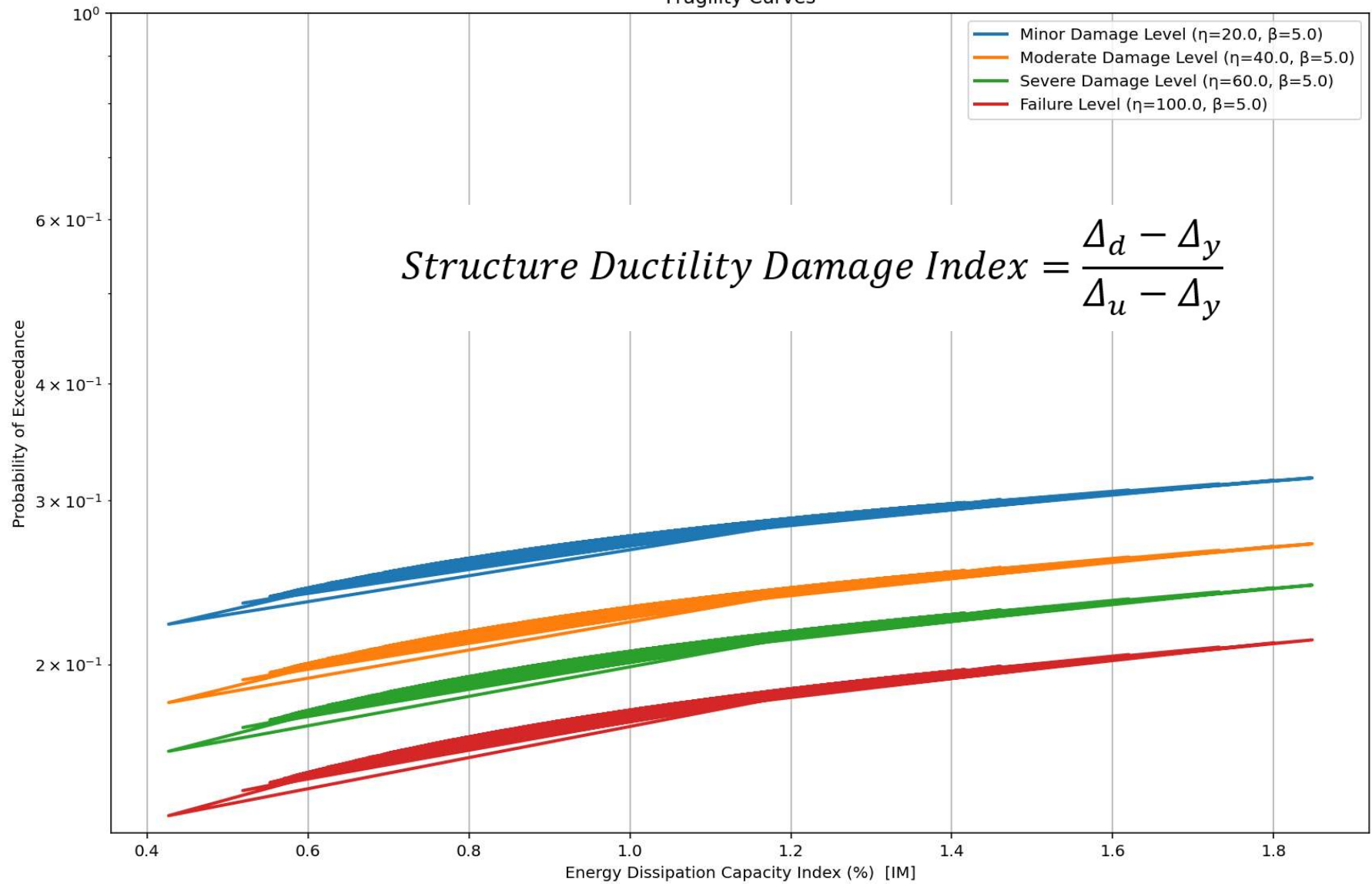




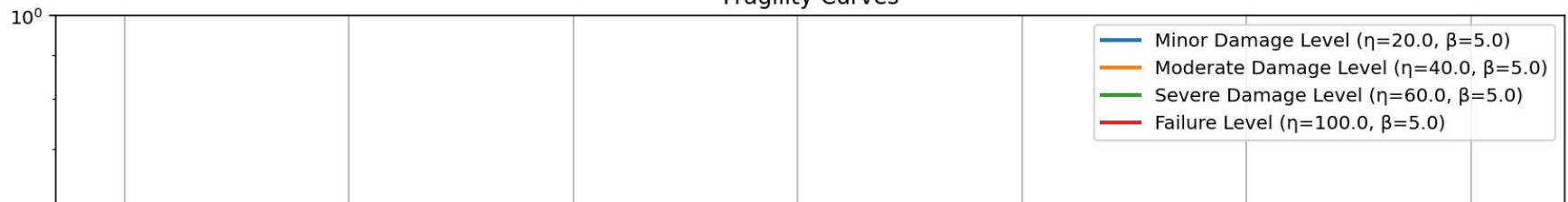




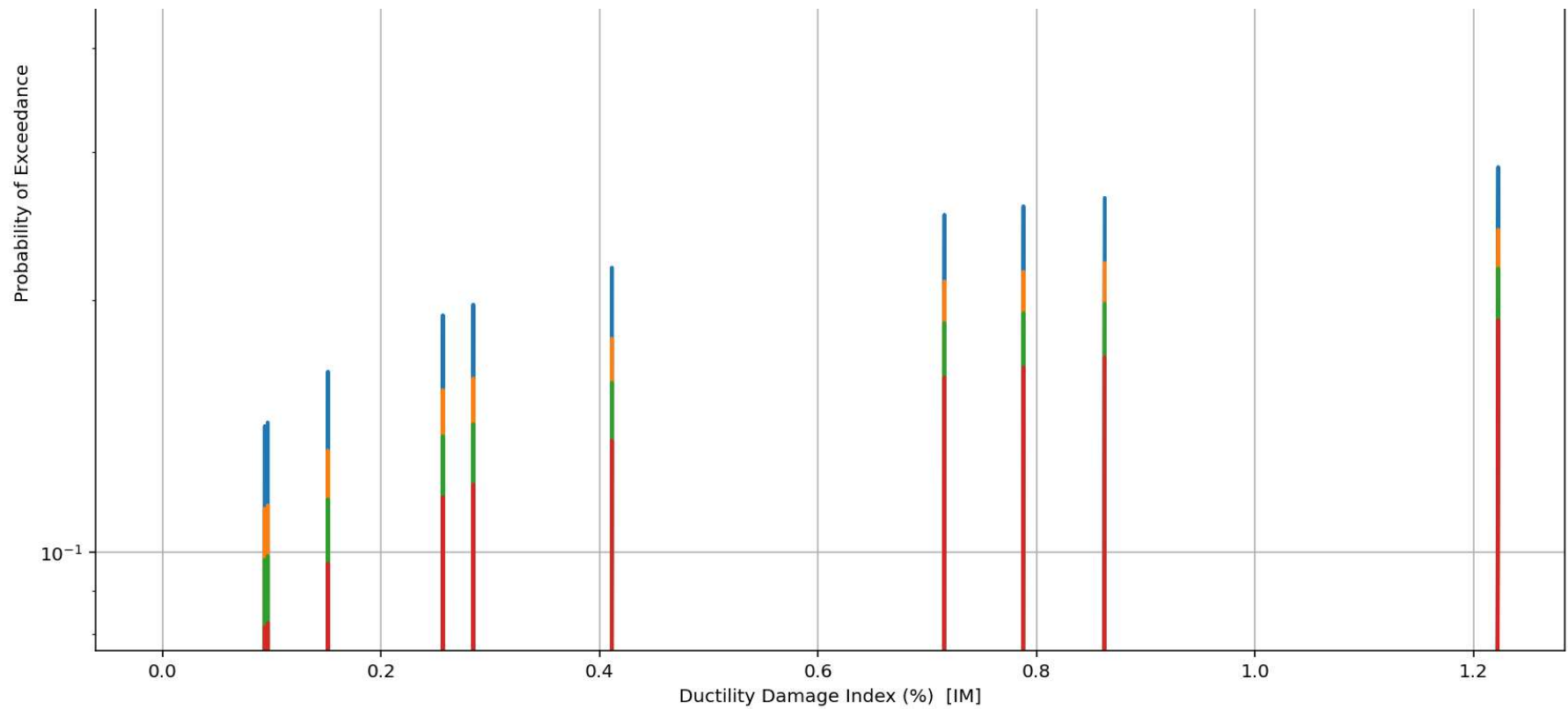
Fragility Curves



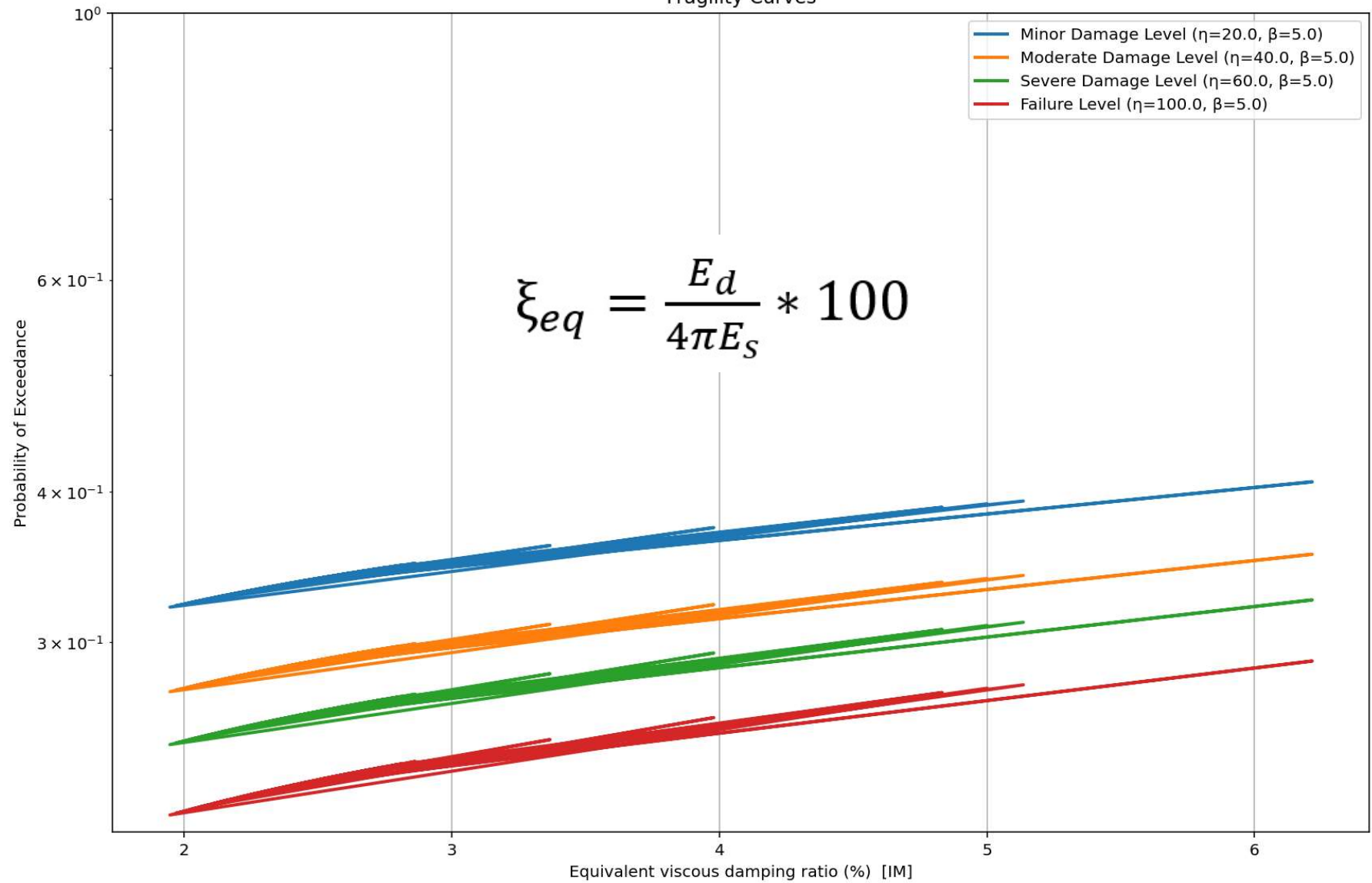
Fragility Curves



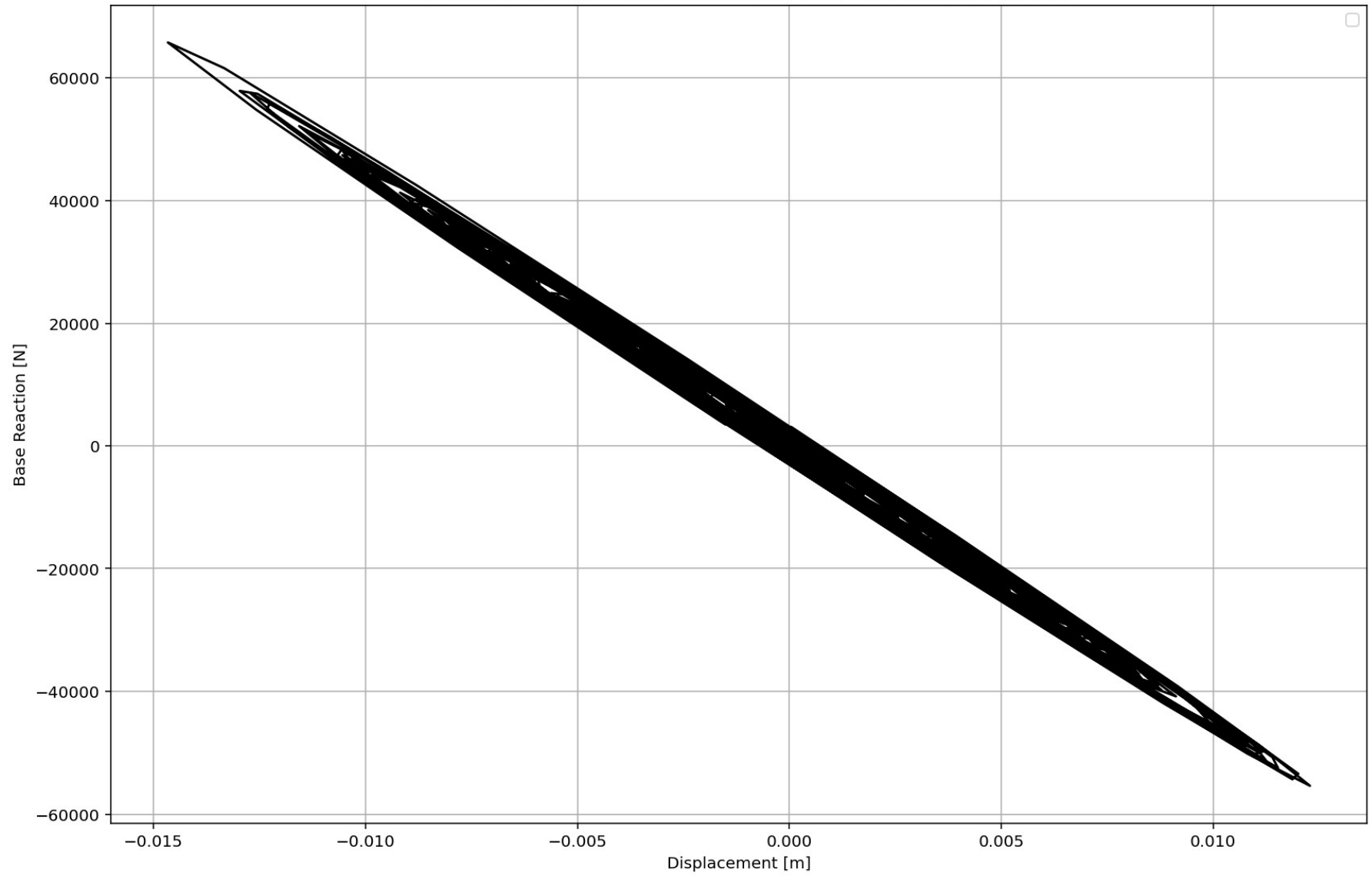
$$\text{Energy Dissipation Capacity Index} = \frac{E_d(\text{Dissipated Energy from Earthquake})}{E_d(\text{Dissipated Energy from Cyclic Displacement})} * 100$$

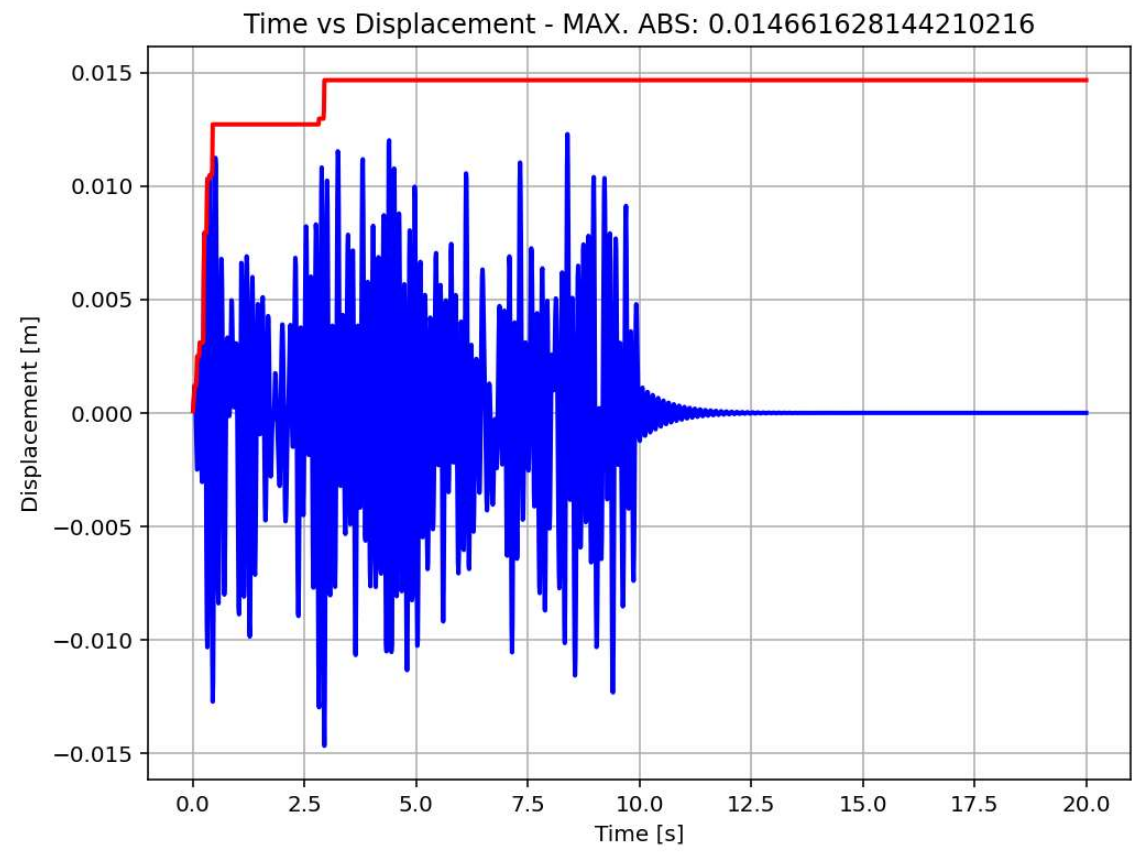


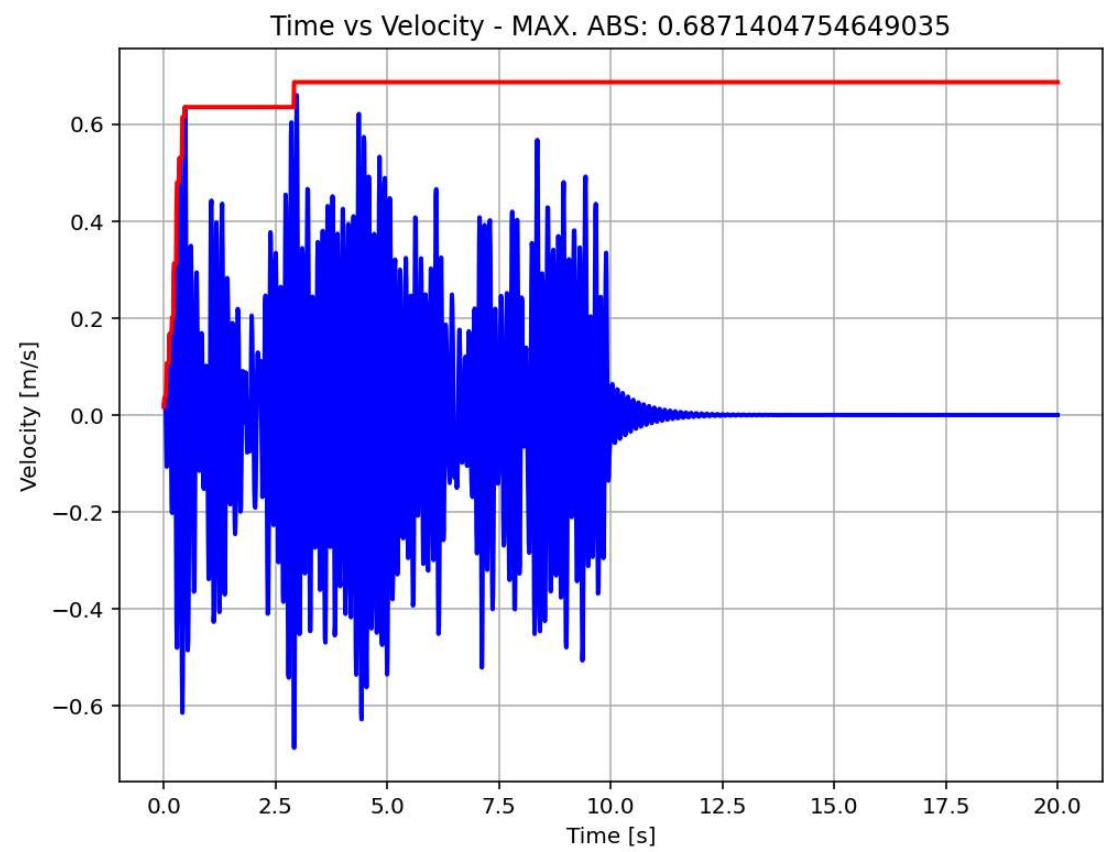
Fragility Curves

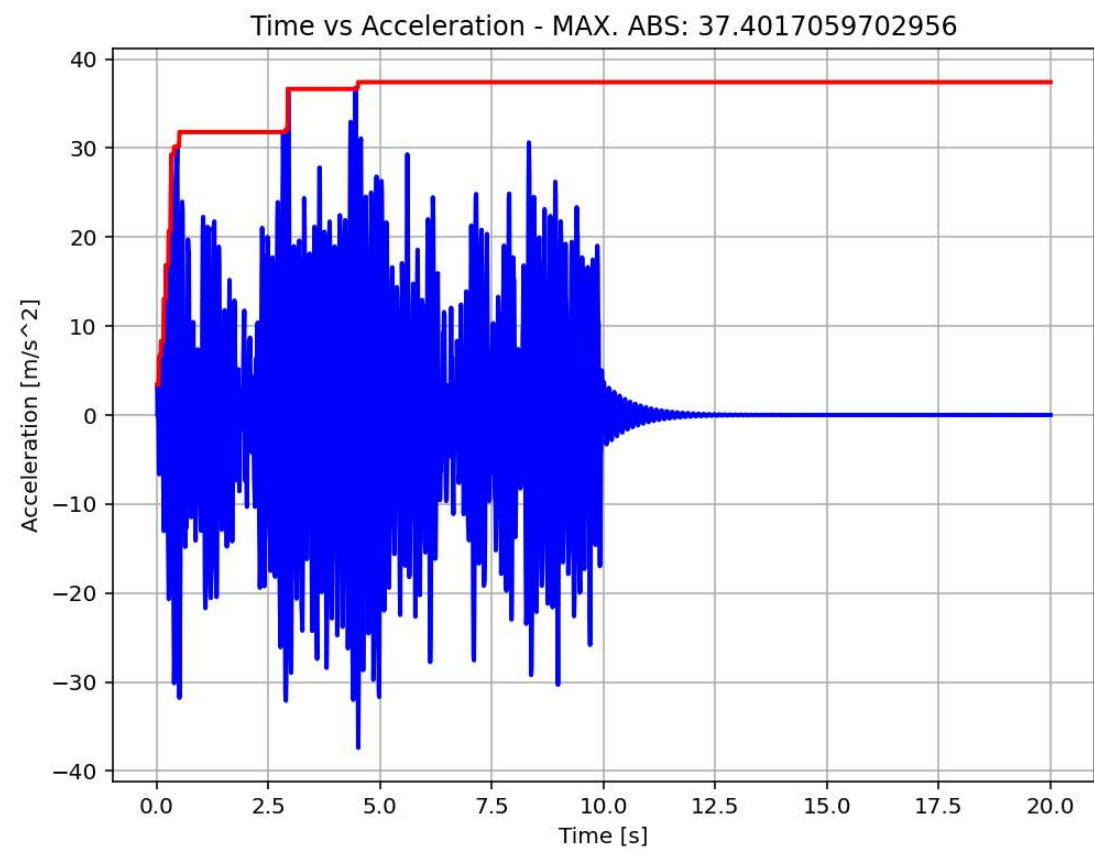


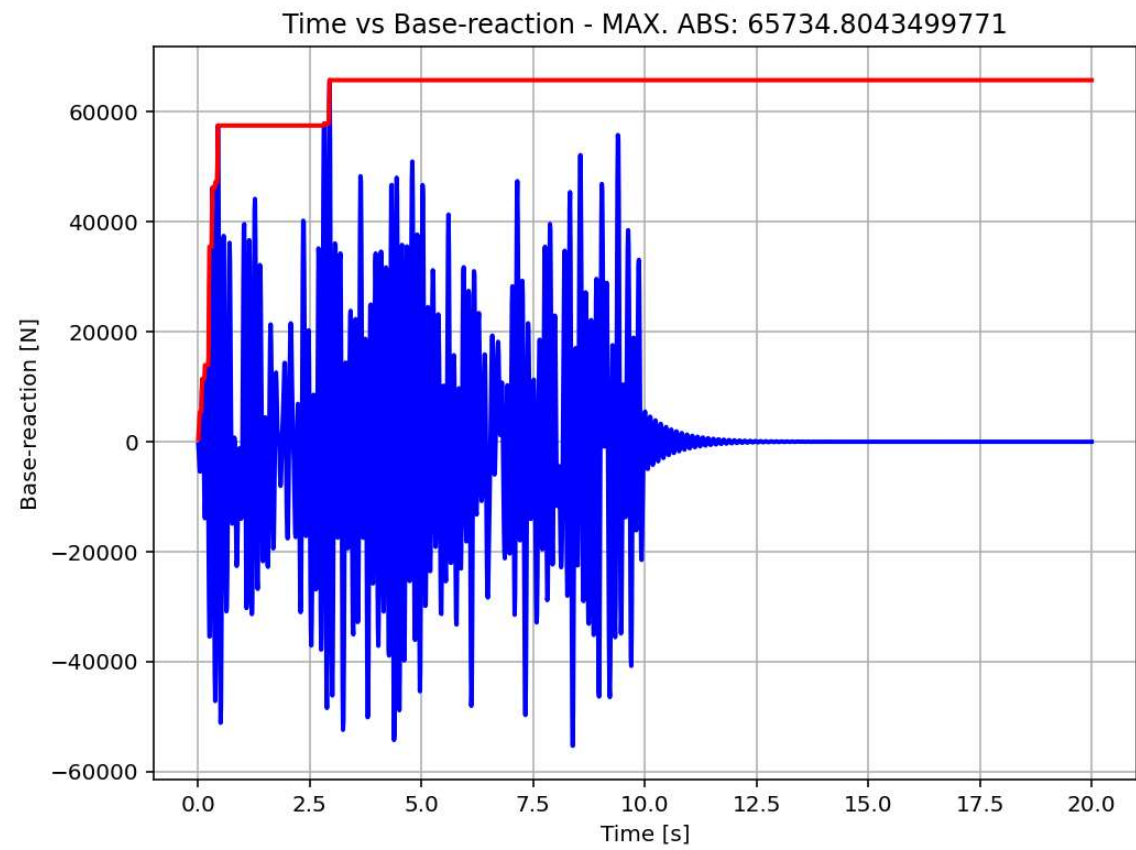
Displacement & Base Reaction Relation From Last Dynamic Analysis

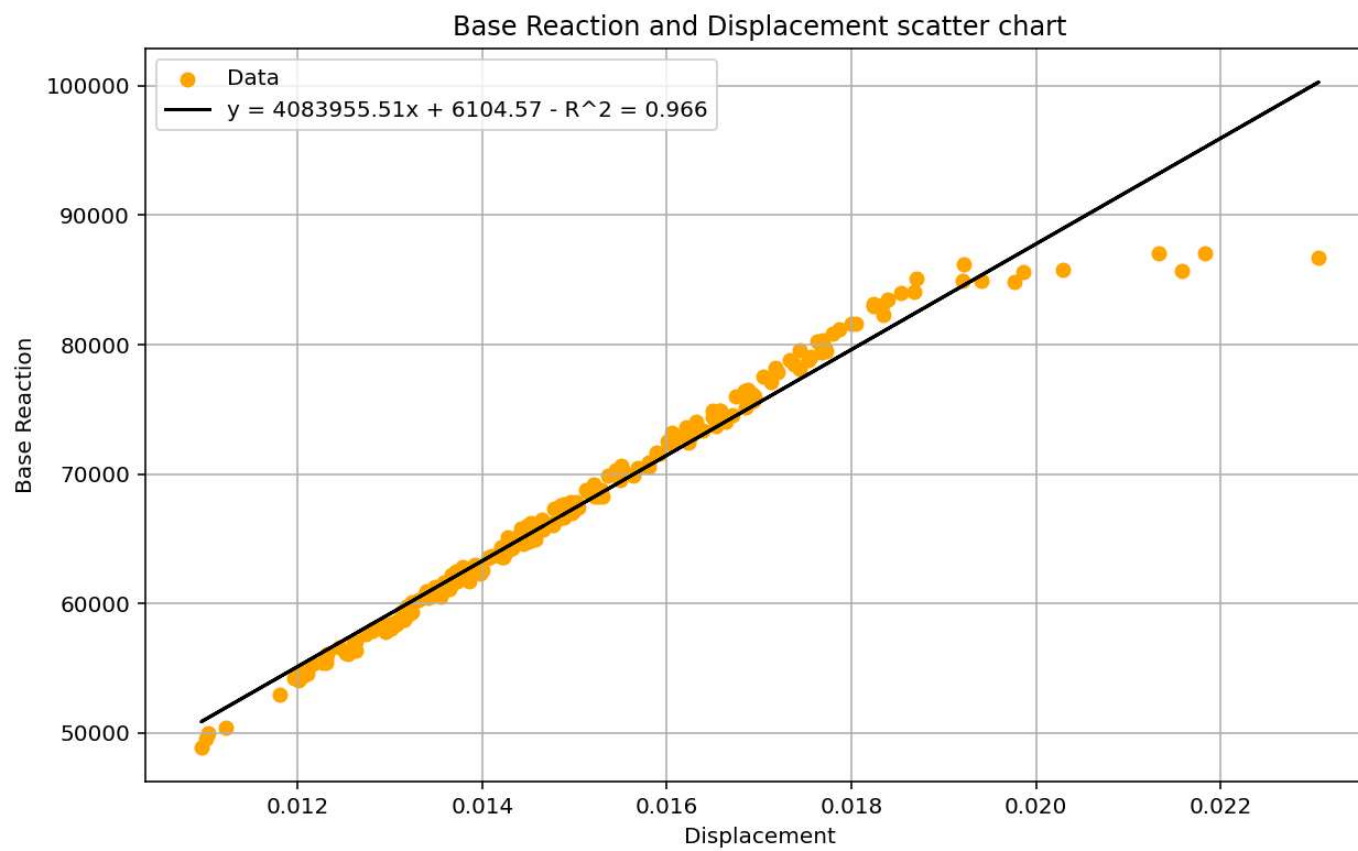


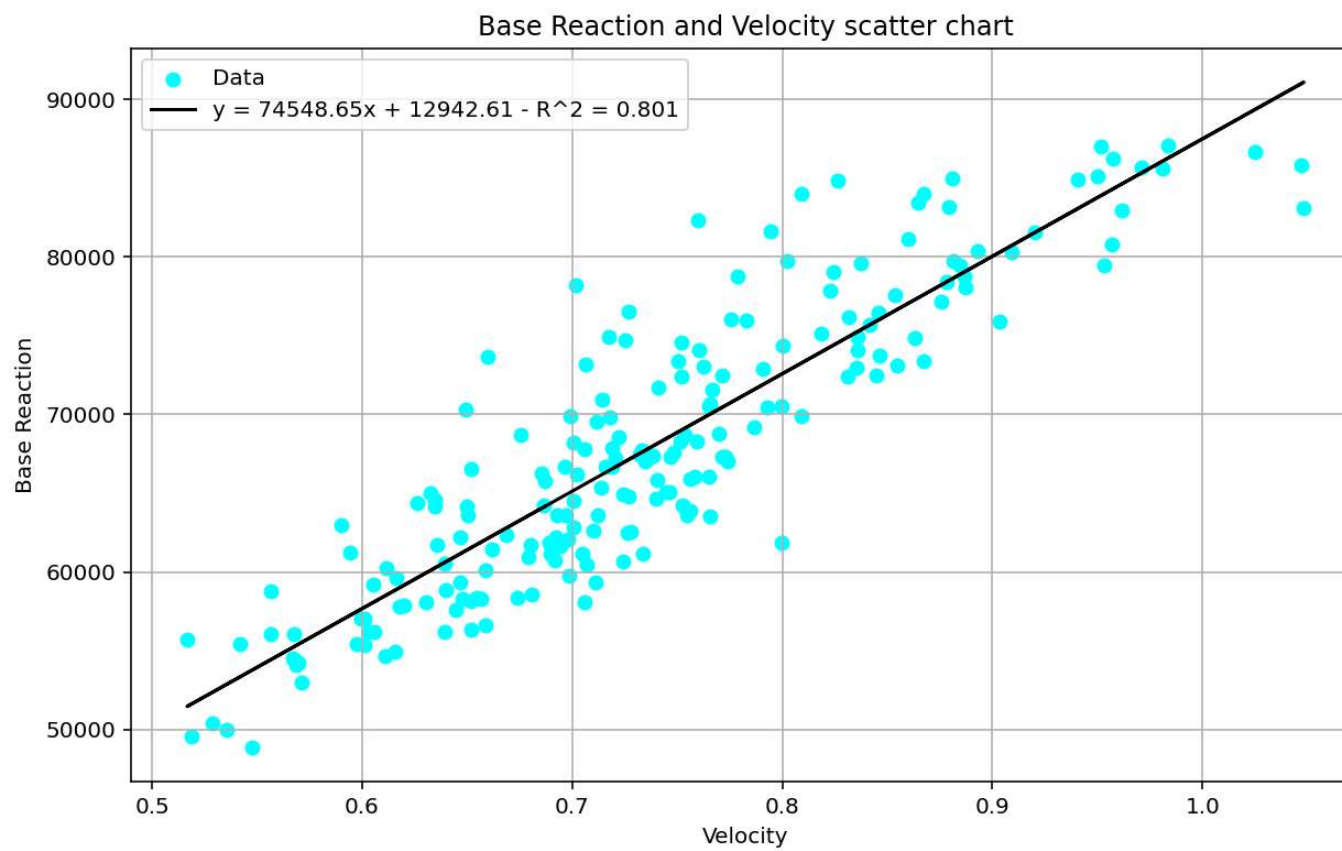


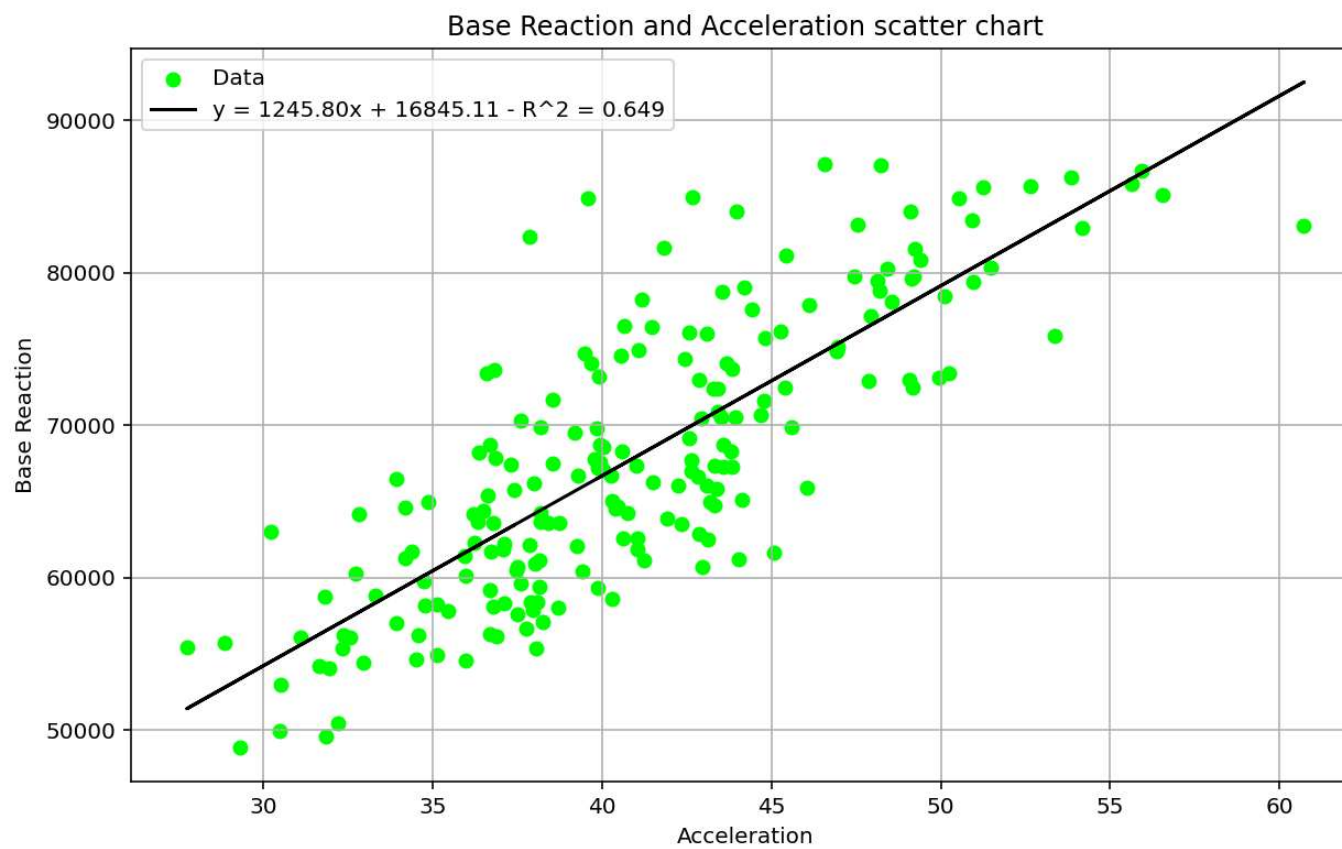


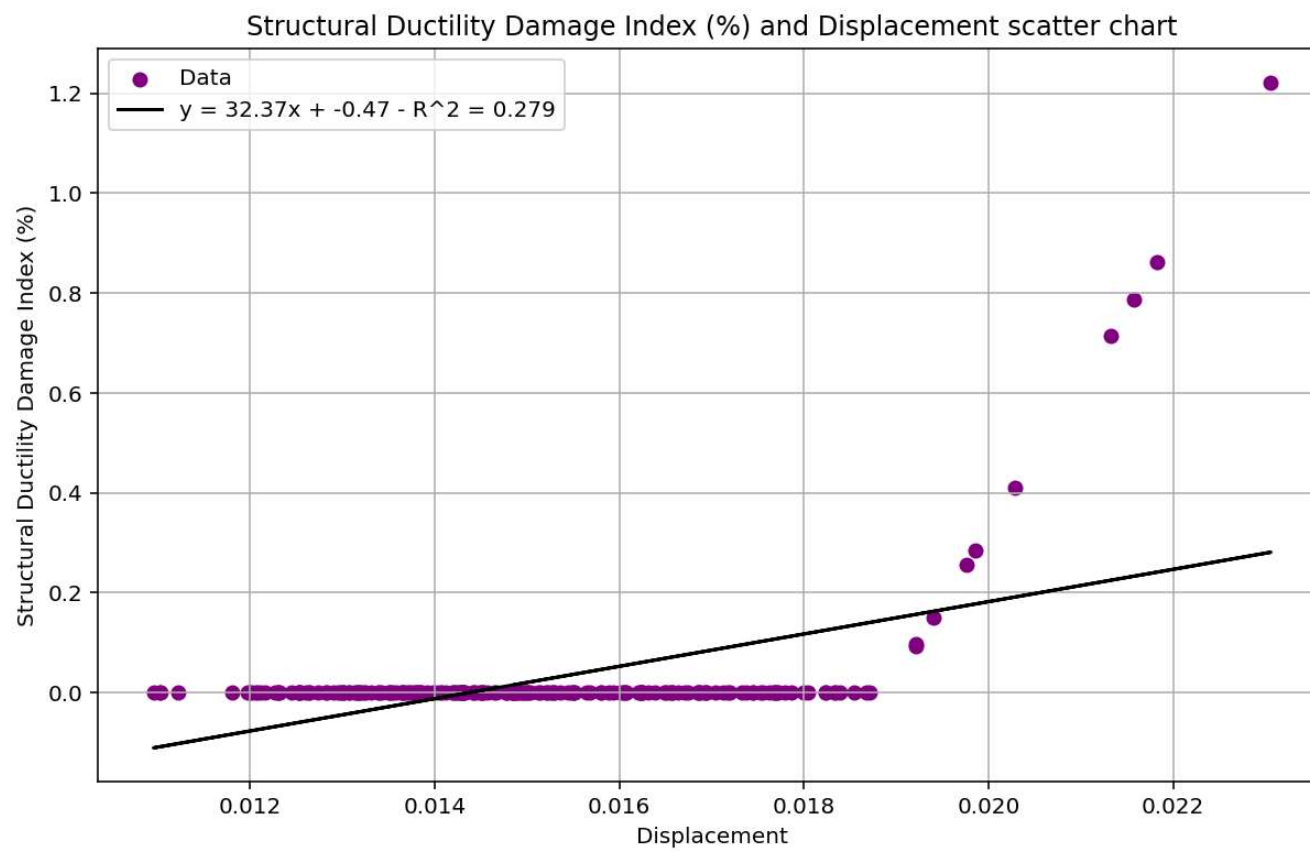


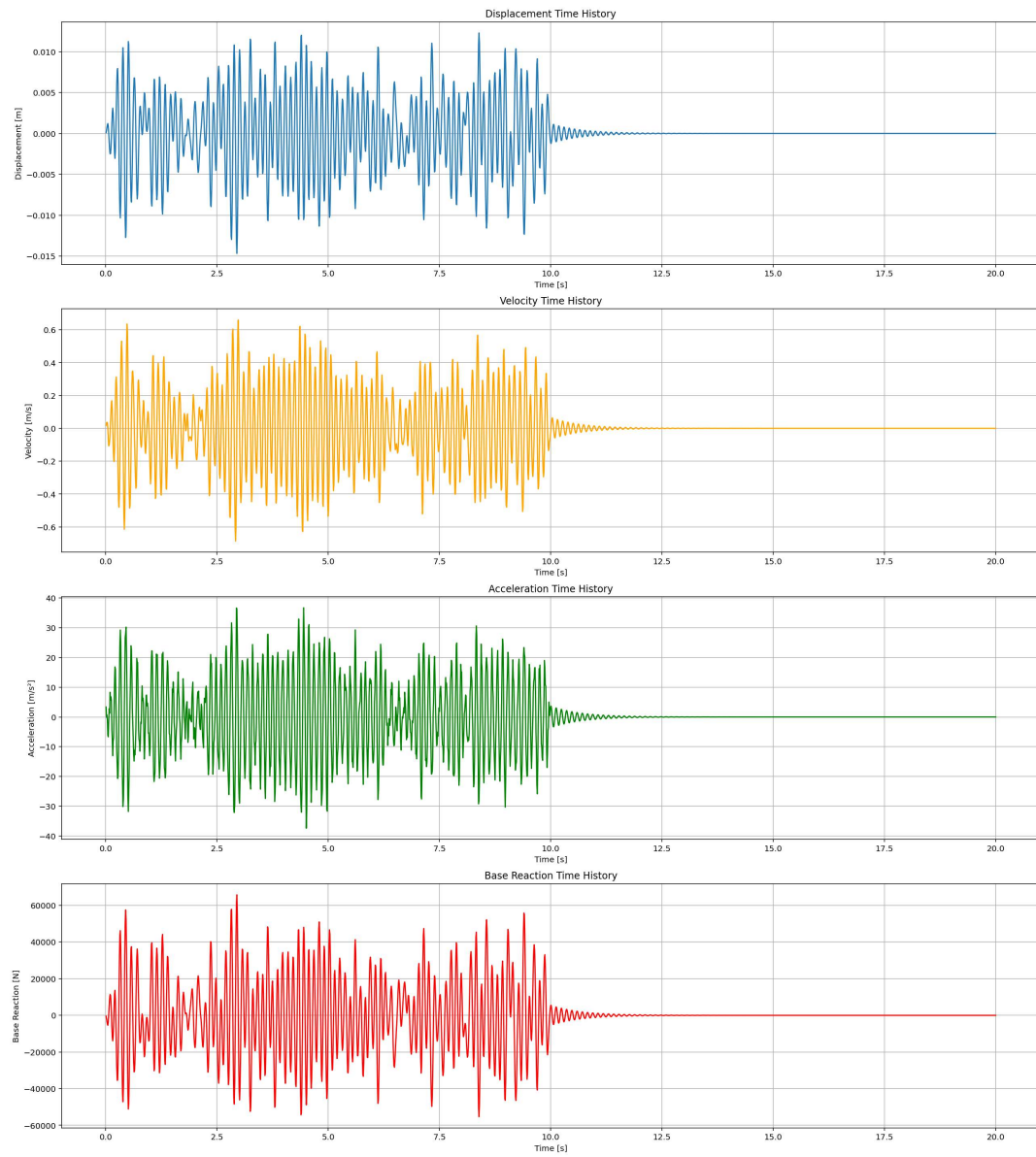


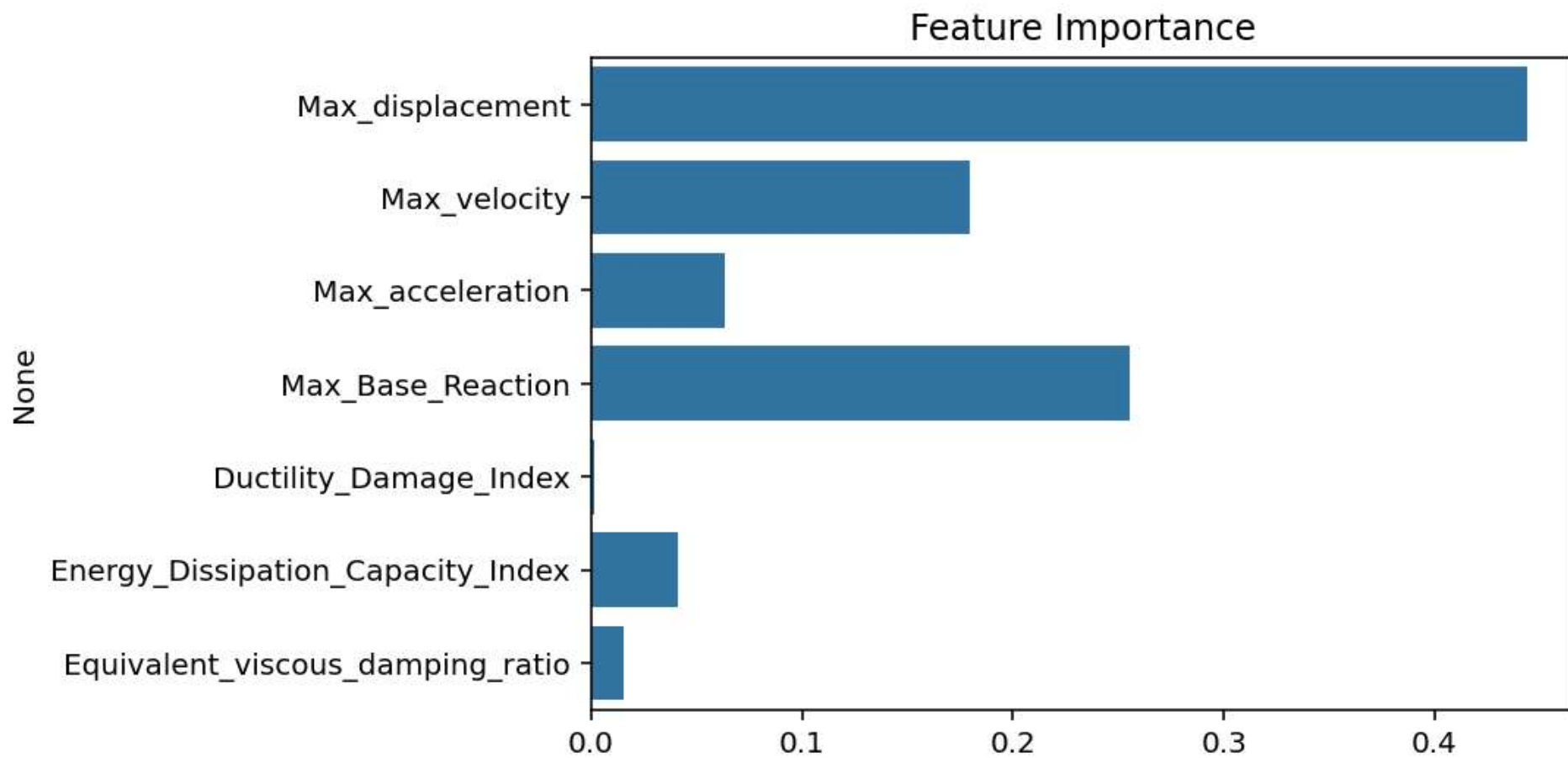












Correlation Heatmap

